

PRICING AGRICULTURAL ENVIRONMENTAL SERVICES UNDER POLICY-CONSISTENT NORMALIZATIONS

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Abstract

Conservation programs differ in how they reward farmers: practice-based programs compensate for adopting management practices, while outcome-based programs pay for measured environmental improvements. Yet the private costs of compliance under these alternative designs remain unclear. We develop a duality-based framework to value agricultural environmental services under policy-consistent normalizations. A normalization selects a numeraire that reflects program objectives and determines how trade-offs among market and environmental outcomes are evaluated. Using Illinois corn production data, we demonstrate that practice-based normalizations result in lower shadow prices for carbon sequestration and greenhouse gas abatement, whereas outcome-based normalizations yield higher costs. Erosion control shows limited variation. By establishing shadow prices as policy-dependent measures, the framework provides a consistent economic basis for comparing conservation programs and setting incentive payments that align private management with environmental goals.

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1 Introduction

This paper addresses a fundamental problem in agricultural and environmental policy: how to assign consistent economic value to environmental services and crops across different policy objectives. Agriculture produces market goods and nonmarket environmental outcomes. Corn is produced alongside carbon sequestration, greenhouse gas emissions, and soil erosion, which lack market prices. Governments attempt to manage these outcomes through conservation programs that influence land-use and production decisions. The scale of these efforts is substantial. U.S. conservation spending alone exceeds \$5 billion per year (ERS, 2025), and global agricultural support totals hundreds of billions annually (Laborde et al., 2021; OECD, 2025). Yet existing valuation methods seldom align with the structure of these programs and the private costs of compliance, limiting their usefulness for evaluating efficiency or setting priorities. This paper develops a framework that bridges this gap by linking valuation to program design.

Conservation programs differ in how they define and reward environmental performance. Practice-based programs compensate farmers for specific management practices, such as reduced tillage or nutrient management, while outcome-based programs link payments to measured improvements in carbon storage, emissions, or erosion reduction. However, the private costs of compliance with these programs are unknown, creating a challenge for both program design and evaluation. Current methods rely on coarse benefit-cost analyses or model-based damage estimates that fail to capture the interaction between production decisions and environmental outcomes (Hosseini-Yekani et al., 2024). Without a consistent valuation framework that accounts for these interdependencies, comparisons across policy instruments or assessments of conservation efficiency remain unreliable.

This paper develops a framework to value agricultural environmental services under policy-consistent normalizations. Because conservation programs involve environmental outcomes that lack market prices, we use shadow prices to quantify the private trade-offs farmers face when improving environmental performance. Shadow prices are derived as trade-offs between netputs from the dual representation of the production technology.¹ The direction and point at which these trade-offs are measured affect the resulting prices. A normalization selects this direction and point. Earlier work in production economics has treated normalization as a mathematical convention that fixes the orientation or scaling of directional distance functions rather than as an economically meaningful choice (Färe et al., 1985; Chambers & Färe, 2020; Chambers, 2025). We demonstrate that the choice of numeraire embeds a policy perspective, determining which production adjustments—input reduction, output expansion, or environmental improvement—are valued.

We map normalizations to policy approaches. We consider practice-based programs that emphasize input reduction, output expansion, or both, and outcome-based programs that reward

¹ *Netputs* refer to a production plan or vector of inputs and outputs.

environmental improvements. The model incorporates both beneficial and adverse environmental outcomes within the production set and represents the technology nonparametrically to remain close to observed data. This approach avoids reliance on restrictive functional forms (Zhou et al., 2014), ensures that shadow prices reflect genuine trade-offs (Podinovski & Førsund, 2010; Podinovski, 2019), and provides a transparent basis for assessing how valuations respond to changes in conservation priorities.

Our empirical analysis uses field-level data on corn production in Illinois for 2023 from Precision Conservation Management (PCM). The dataset integrates detailed records of management practices, production, and input use with environmental metrics generated by Cool Farm Tool (CFT) and Field to Market (FTM). CFT provides estimates of carbon sequestration and greenhouse gas (GHG) emissions, while FTM provides soil loss due to water and wind erosion. These combined data enable us to quantify the joint production relationships between crop output and environmental outcomes. Illinois provides a suitable setting because it encompasses a broad range of soil quality, slope, and management intensity within a single policy environment. The state also accounts for a significant share of U.S. corn production, making the findings relevant for national conservation programs.

We find that normalization matters for the derived value of environmental services. Normalizations aligned with practice-based incentive programs yield lower shadow price estimates than those tied to outcome-based programs. The contrast is strongest for carbon sequestration and greenhouse gas emissions, where alternative normalizations result in significant differences in implied costs and benefits. Erosion control exhibits limited variation, suggesting that soil conservation is less sensitive to the policy framework applied. These results demonstrate that shadow prices are not fixed features of the production system, but rather reflect the policy orientation embedded in the normalization process. Making this dependence explicit is essential for credible valuation and for comparing conservation programs on consistent economic terms.

This paper contributes to two strands of literature. First, it addresses environmental policy design by developing a framework to value the private costs of compliance under alternative regulatory and incentive approaches. Environmental policies may define multiple pathways to achieve objectives, whether through input restrictions or outcome-based performance measures, but lack consistent methods to compare these alternatives from the regulated entity’s perspective. Our framework derives the costs of compliance from observed production behavior, accounting for joint production of crops and environmental outcomes. In agriculture, this framework enables comparison of practice-based programs, such as the Environmental Quality Incentives Program (EQIP) and the Conservation Stewardship Program (CSP), which compensate farmers for adopting specified practices (Claassen & Ribaud, 2016; Hellerstein, 2017), with outcome-based programs that tie payments to measured environmental performance (Burton & Schwarz, 2013; Lapierre et al., 2023). Research on targeting demonstrates the value of prioritizing land and practices based on expected

ecological benefits (Roberts & Bucholtz, 2005; Lubowski et al., 2006). Yet, most evaluations rely on program-specific scoring systems or modeled biophysical indicators rather than economic measures derived from production behavior. By deriving shadow prices from field-level data under alternative normalizations, we provide an internally consistent basis for comparing private compliance costs across policy instruments.

Second, this work advances environmental and production economics by reinterpreting shadow prices as policy-dependent measures. Existing approaches to valuation of environmental outcomes include hedonic pricing models that infer willingness to pay from property values (Greenstone & Gallagher, 2008; Keiser & Shapiro, 2019), avoided damage assessments that estimate benefits through cost savings (Keeler et al., 2012; Sobota et al., 2015), integrated assessment models that connect environmental outcomes to economic welfare (Nordhaus, 2017; Pindyck, 2017), and stated preference approaches such as contingent valuation and choice experiments (Johnston et al., 2017). While these approaches have been central to ecosystem service valuation, they operate at coarse spatial scales, rely on external damage estimates or elicited preferences, and treat production and environmental outcomes as separate processes. We complement this literature by modeling the technology set that produces market goods and environmental outcomes, then deriving private shadow prices through the dual representation of technology. This approach reveals the private trade-offs producers face—what they forgo in market revenue or extra costs they pay when improving environmental performance.

The framework builds on the production economics literature using both econometric approaches, such as stochastic frontier analysis (Aigner et al., 1977; Kumbhakar et al., 2015; Choi et al., 2023), and mathematical programming methods, including data envelopment analysis (Charnes et al., 1978; Cook & Seiford, 2009). A notable issue in this literature is the orientation of efficiency measurement. These different orientations reflect different normalization criteria measuring distance to frontiers and for calculating efficient reference points on which these distances are based. Different functions can represent the technology, and they can be input-oriented, output-oriented (Banker et al., 1984), hyperbolic (Färe et al., 1985), or endogenous (Färe et al., 2013; Ray, 2007).² Most studies fix a single evaluation direction, which in turn determines the normalization used to measure trade-offs. We instead develop a menu of normalizations corresponding to alternative policy approaches. The set of normalizations parallels the role of distance functions, each defining a reference direction for evaluating deviations from the efficient frontier. We build on advances in conjugate duality (Chambers, 2025), demonstrating that valuations vary systematically with the policy perspective applied.

The paper most closely related to ours is Kunwar et al. (2025), which derives environmental shadow prices for carbon services in agriculture using a by-production framework and land as a

²Hyperbolic or radial distance functions are a special case of the more general directional distance functions (Chambers & Färe, 2020)

numeraire. The study provides the first empirical evidence on the private value of carbon sequestration and greenhouse gas mitigation. While [Kunwar et al. \(2025\)](#) fixes land as the numeraire—a choice reflecting production technology but not policy design, we introduce policy-relevant normalizations. Conservation programs target management practices (such as cover cropping and minimum tillage), or reward measurable environmental outcomes (such as carbon storage or emission reductions). Our normalizations mirror these policy approaches, enabling direct comparison of private compliance costs across program types.

Together, these elements position the paper at the intersection of production theory and conservation policy design, offering a framework that ties valuation to incentive structures. [Section 2](#) develops the theoretical framework and explains how normalization choices shape valuation. [Section 3](#) describes the data. [Section 4](#) details how policy programs correspond to alternative normalizations. [Section 5](#) presents results and discusses implications for conservation design. [Section 6](#) concludes.

2 Methods

2.1 The Technology

Our approximation of corn production technology T is based on a set of I corn fields with a K -dimensional input vector x , a primary output corn y , a beneficial environmental output carbon sequestration s , and an M -dimensional adverse environmental output vector b . The technology set is given by

$$T = \{(x, y, s, b) \in \mathbb{R}_+^{K+1+1+M=S} : x \text{ can produce } (y, s, b)\} \quad (1)$$

We denote a vector of netputs by z . We adopt a subset of the standard axioms of the neo-classical production model outlined below³:

A.1. T is nonempty: $T \neq \emptyset$

A.2. There is no free lunch: Suppose $z = (0, y, s, b)$ and $y, s, b > 0$ then $z \notin T$.

A.3. T is a closed set.

A.4. T is a convex set: If z^0 and $z^1 \in T$ then $z^0 + \lambda(z^1 - z^0) \in T$ for all $\lambda \in [0, 1]$.

³For more details on the standard axioms of production theory, refer to [Chambers et al. \(1998\)](#) and [Färe & Grosskopf \(2005\)](#).

2.2 Empirical Technology

We specify the corn production technology with seven inputs and five outputs. We present the net-puts, their notations, and units of measurement in Table 1. The primary output is corn production y , and beneficial environmental output is carbon sequestration s . Adverse environmental outputs are GHG emissions b_1 , water erosion b_2 , and wind erosion b_3 .

Corn production, measured in bushels, is the primary output of agricultural production technology. Carbon sequestration, measured in tons of carbon dioxide equivalent (tCO₂ eq.), is the amount of CO₂ captured and stored in the soil. GHG emissions in corn production include CO₂ and nitrous oxide (N₂O) emissions.⁴ They derive from fuel combustion in machinery, fertilizer application, and other agricultural practices. We consider two types of soil erosion: water erosion and wind erosion. Water erosion measures the amount of soil lost in tons due to water runoff during heavy rainfall or irrigation. It washes out essential nutrients from the field and pollutes nearby water resources (Pimentel et al., 1995; Trimble & Crosson, 2000). Wind erosion quantifies the soil lost in tons due to wind action, which affects air quality.

Land is a fundamental input in corn production. Larger fields support higher grain production by facilitating mechanized farming and enhancing operational efficiency (MacDonald, 2020; Sheng & Chancellor, 2019).⁵ Land also impacts environmental outcomes. Larger fields can sequester more carbon dioxide in the soil and plant biomass (Paustian et al., 2016). However, larger fields can also increase GHG emissions through the use of machinery and inputs. On sloped terrain, larger fields are at a higher risk of water and wind erosion without adequate conservation practices in place.

The Soil Productivity Index (SPI) evaluates the productivity potential of various soil types in supporting crop growth under optimal management conditions. In Illinois, SPI values for agricultural land range from 100 to 147, with higher values indicating better potential. These values are derived from a comparative analysis against a baseline soil type—Muscatine silt loam, one of the most productive soils in Illinois (Olson, 2000).

Non-land costs include direct costs (seeds, fertilizers, and pesticides), power costs (energy used for mechanized operations), and overhead costs. These inputs directly affect environmental outputs. Nitrogen fertilizers generate N₂O emissions. Mechanized fieldwork releases CO₂ through fuel combustion and N₂O and CO₂ through tillage. Intensive tillage disrupts soil structure, increasing susceptibility to water and wind erosion.

The Soil Erodibility Factor (K-factor) measures a soil’s susceptibility to water and erosion. It ranges from 0.02 to 0.69 in Illinois soils (Goldman et al., 1986; Mitchell & Bubenzer, 1980). A

⁴We do not consider CH₄ emissions, as our fields do not use compost or manure, a primary source of CH₄ (Thangarajan et al., 2013).

⁵However, the relationship between farm size and productivity has long been a topic of debate, and researchers have found an inverse relationship in developing countries (Barrett, 1996; Barrett et al., 2010).

Table 1: Netputs in the corn production technology.

Input - (Unit)	Output - (Unit)
(x_1) Land (acres)	(y) Grain production (bushels)
(x_2) Soil productivity index (49 – 147)	(s) Carbon sequestration (tCO ₂ eq.)
(x_3) Non-land cost (\$)	(b_1) GHG emissions (tCO ₂ eq.)
(x_4) Soil erodibility factor (0.02 – 0.69)	(b_2) Water erosion (tons of soil)
(x_5) Field slope (%)	(b_3) Wind erosion (tons of soil)
(x_6) Precipitation (millimeter)	
(x_7) Simple daily intensity index (SDII; millimeters per day)	

Note: Parentheses before and after the netput description provide notation and units of measure.

higher K-factor, indicating greater susceptibility to erosion, is determined by soil properties such as texture, organic matter content, structure, and slope. Field slope increases erosion risk. Steeper slopes accelerate water flow, detaching and transporting soil particles. This increased runoff can intensify soil loss in fields lacking vegetation or erosion control measures. Intensive tillage on sloped land can destabilize soil structure, increasing vulnerability to both water and wind erosion. Consequently, high-K soils may require additional management practices, such as cover cropping or conservation tillage, to limit soil degradation.

Precipitation affects productivity and environmental outcomes.⁶ Total precipitation determines water availability for crop growth and biomass accumulation, influencing yield and carbon sequestration. The Simple Daily Intensity Index (SDII) measures average precipitation intensity on wet days (Dollan et al., 2022; Yu et al., 2020). It is calculated as total precipitation over a period divided by the number of wet days. SDII captures whether precipitation arrives as gentle, frequent events or intense storms, which affects erosion, runoff, and nutrient dynamics. While sufficient precipitation supports carbon sequestration through increased biomass production, intense rainfall can wash away organic material, thereby reducing the carbon storage potential.

Precipitation also impacts GHG emissions. Excessive precipitation in poorly drained soils creates waterlogged conditions, leading to anaerobic environments that favor denitrification,⁷ releasing N₂O. High-intensity rainfall leads to surface runoff, nutrient leaching, and soil erosion (Pimentel et al., 1995).

Inputs, such as land area x_1 and SPI x_2 , can be increased indefinitely without a negative effect on the primary output, suggesting a non-decreasing marginal product. Non-land costs, which include fertilizers and pesticide costs, can have a potential congestion effect (Corteva, 2023; Vargas et al., 2015). Other inputs, such as the soil erodibility factor, field slope, and weather variables (x_6 and

⁶We exclude temperature as a conditioning weather variable because, in our 2023 Illinois sample, growing-season temperature exhibits little cross-sectional variation across fields. In a single state and year, such limited variation adds little identifying information, so we focus on precipitation-based metrics.

⁷Denitrification in the soil is a microbial process that occurs under anaerobic or low-oxygen conditions involving the conversion of nitrate (NO₃⁻) into gaseous forms of nitrogen (N₂O and N₂) (Davidson & Kanter, 2014).

x_7), have detrimental effects if they reach excessive levels (USDA, 2022; Yoksoulian, 2019). Given these dynamics, we assume x_1 and x_2 are freely disposable and x_3 to x_7 are not. Let x^1 denote our vector of freely disposable inputs (x_1, x_2) and x^2 denote our vector of non-freely disposable $(x_3, x_4, x_5, x_6, x_7)$. We assume primary outputs y to be freely disposable, as farmers can always produce below the maximum feasible yield by harvesting less through inefficient operations.

Our beneficial environmental outputs, carbon sequestration s_1 , and biodiversity s_2 are intrinsically linked to the primary output y . Increased grain production indicates greater crop biomass, which enhances carbon sequestration by storing more carbon in the soil and plant (Dixit et al., 2023). In addition, carbon sequestration is influenced by soil productivity, erodibility, field slope, field management practices, and weather variables. Due to these interdependencies, one cannot reduce carbon sequestration independently of other netputs, and we assume these beneficial environmental outputs are not freely disposable.

Altering adverse environmental outputs b often requires adjustments in resource use, management practices, or production intensity, which can impose costs or lead to trade-offs in production. For example, lowering (increasing) GHG emissions involves reducing (increasing) nitrogen fertilizer use (Menegat et al., 2022), and reducing erosion requires additional soil conservation efforts that redirect resources from production activities (Silva et al., 2024). Thus, we assume our adverse environmental variables are not freely disposable.

2.3 Recession Cone of T

We use the notion of recession cones to discuss the netputs' disposability properties following Chambers (2025). In general, the recession cone T_∞ of T can be defined as the set of all directions $d \in \mathbb{R}^S$ such that for every point $z \in T$, the ray $z + \beta d$ for $\beta \geq 0$ remains within T . Mathematically,

$$T_\infty = \{d \in \mathbb{R}^S : (z + \beta d) \subset T, \beta \geq 0\}. \quad (2)$$

We follow Farrell (1957) and Afriat (1972) and use netput observations together with our axioms to approximate the technology. Let $\mathcal{D}$ denote the set of I corn fields represented by a $(K^1 + K^2)$ - dimensional input vector and a $(1 + 1 + M)$ - dimensional output vector. Mathematically,

$$\mathcal{D} = \{(x_i^1, x_i^2, y_i, s_i, b_i) \in \mathbb{R}^{K^1 + K^2 + 1 + 1 + M = S}; i = 1, \dots, I\} \quad (3)$$

The smallest set (*convex hull*) of $(x, y, s, b) \in \mathbb{R}^S$ satisfying A.4. is given by

$$\text{co}\{T^{\mathcal{D}}\} = \left\{ \begin{array}{l} (x^1, x^2, y, s, b) : x_k^1 = \sum_i \gamma_i x_{ik}^1, x_k^2 = \sum_i \gamma_i x_{ik}^2, y = \sum_i \gamma_i y_i, \\ s = \sum_i \gamma_i s_i, b_m = \sum_i \gamma_i b_{im}, \gamma_i \geq 0 \text{ \& } \sum_i \gamma_i = 1 \end{array} \right\} \quad (4)$$

where $\text{co}\{\}$ denotes the convex hull and γ_i the weight that defines the convex combination of the fields. The constraint $\sum_i \gamma_i = 1$ reflects convexity. The recession cone of $T^{\mathcal{D}}$ in (4) is a singleton set and given by $T_{\infty} = \{0\}$, implying that T is compact and is Charnes et al. (1985) *empirical production sets*.

Following the discussion about the interaction between netputs in the corn production technology in Section 2.2, we redefine the recession cone T_{∞} as⁸

$$T_{\infty} = \{(d_{x^1}, 0, d_y, 0, 0) \in \mathbb{R}_+^{K^1} \times \mathbb{R}^{K^2} \times \mathbb{R}_- \times \mathbb{R} \times \mathbb{R}^M\} \quad (5)$$

Using the production-theoretic notion of netput disposability, this implies that x^1 and y are freely disposable, while x^2 , s , and b are not. Directions for x^2 , s , and b are set to 0, which means that they cannot be moved in any direction infinitely given other netputs fixed. Thus, our empirical technology consistent with T_{∞} in (5) is defined as $T^{\mathcal{D}}$, and its polyhedral approximation is given by

$$T^{\mathcal{D}} = \left\{ \begin{array}{l} (x^1, x^2, y, s, b) : \\ x_k^1 \geq \sum_i \gamma_i x_{ik}^1, x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ y \leq \sum_i \gamma_i y_i, s = \sum_i \gamma_i s_i, \\ b_m = \sum_i \gamma_i b_{im}, \gamma_i \geq 0 \text{ \& } \sum_i \gamma_i = 1 \end{array} \right\}. \quad (6)$$

2.4 Efficiency Measure

The frontier is characterized by the best-practice combinations of netputs (*best-practice frontier*) of $T^{\mathcal{D}}$. Each field's distance to the best-practice frontier indicates its efficiency, with the measures being dependent on the normalization Q – a set of price vectors that defines the numeraire. The set of prices $q \in Q$ defines the direction we measure distance (e.g., toward more corn output, less nitrogen input, etc.).

Following Chambers (2025), for any netput vector $z \in \mathbb{R}^S$, we define the Pareto efficiency

⁸The direction $d_y \in \mathbb{R}_-$ implies that y could be negative. However, it is clear that y is constrained to be non-negative from the definition of T in (1) and the recession cone in (2). For brevity, we do not repeat the constraints defined in T_{∞} each time we define T_{∞} .

measure, $\delta^Q : \mathbb{R}^S \rightarrow \mathbb{R}$, as

$$\delta^Q(z|T) \equiv \sup\{q'z - \delta^*(q|T) : q \in Q\} = \sup_{q \in \mathbb{R}^{S*}} \{q'z - \delta^*(q|T) - \delta(q|Q)\} \quad (7)$$

where q is a price vector in the normalization set Q , and $q'z$ is the profit of z evaluated at prices q . Expression $\delta^*(q|T)$ denotes the maximum profit achievable in T at prices q and $q'z - \delta^*(q|T)$ is the profit shortfall. The Pareto efficiency measure quantifies the minimum *profit-loss* revealed by directions $q \in Q$.

$$\delta^*(q|T) \equiv \sup\{q'z : z \in T\} = \sup_{z \in \mathbb{R}^S} \{q'z - \delta(z|T)\}. \quad (8)$$

is the support function of T , i.e., the Fenchel conjugate of $\delta(z|T) : \mathbb{R}^S \rightarrow \{0, \infty\}$, the indicator function for T .

$$\delta(z|T) = \begin{cases} 0 & \text{if } z \in T \\ \infty & \text{otherwise} \end{cases}$$

The indicator function enforces that only netputs within T are feasible.

Properties of δ^Q

- (i) Feasible sign: If $z \in T$, then $\delta^Q(z | T) \leq 0$.

Proof: For any $q \in Q$ and $z \in T$, $\delta^*(q | T) = \sup_{w \in T} q'w \geq q'z$. Hence each term $q'z - \delta^*(q | T) - \delta(q | Q) \leq 0$; the supremum over $q \in \mathbb{R}^{S*}$ preserves the inequality.

- (ii) Zero if and only if exposed by some $q \in Q$: If $z \in T$, then $\delta^Q(z | T) = 0 \iff \exists q \in Q : q'z = \delta^*(q | T)$.

Proof: “ $\Rightarrow$ ”: If $\delta^Q(z | T) = 0$, then there exists a maximizing sequence $q_n \in \mathbb{R}^{S*}$ with $\delta(q_n | Q) = 0$, hence $q_n \in Q$, and $q'_n z - \delta^*(q_n | T) - \delta(q_n | Q) \uparrow 0$. By compactness of Q , extract a subsequence $q_n \rightarrow q^* \in Q$. Continuity of $\delta^*(\cdot | T)$ implies $q'^* z = \delta^*(q^* | T)$.

“ $\Leftarrow$ ”: If there exists $q \in Q$ with $q'z = \delta^*(q | T)$, then for this q the term inside the supremum is 0. Hence, the supremum is 0.

- (iii) Monotonicity in T : If $T \subset T'$, then $\delta^Q(z | T) \geq \delta^Q(z | T')$ for all z .

Proof: If $T \subset T'$, then $\delta^*(q | T') = \sup_{w \in T'} q'w \geq \sup_{w \in T} q'w = \delta^*(q | T)$ for all q . Hence $q'z - \delta^*(q | T) - \delta(q | Q) \geq q'z - \delta^*(q | T') - \delta(q | Q)$, and taking the supremum over $q \in \mathbb{R}^{S*}$ yields $\delta^Q(z | T) \geq \delta^Q(z | T')$.

- (iv) Scaling of the normalization: For any $\beta > 0$, $\delta^{\beta Q}(z | T) = \beta \delta^Q(z | T)$.

Proof: By definition, $\delta^{\beta Q}(z \mid T) = \sup_{q \in \mathbb{R}^{S^*}} \{q'z - \delta^*(q \mid T) - \delta(q \mid \beta Q)\}$. Because $\delta(q \mid \beta Q) = \delta\left(\frac{q}{\beta} \mid Q\right)$, set $q = \beta \tilde{q}$ with $\tilde{q} \in \mathbb{R}^{S^*}$. Then $\delta^{\beta Q}(z \mid T) = \sup_{\tilde{q} \in \mathbb{R}^{S^*}} \{\beta \tilde{q}'z - \delta^*(\beta \tilde{q} \mid T) - \delta(\tilde{q} \mid Q)\}$. Because the support function is positively homogeneous, $\delta^*(\beta \tilde{q} \mid T) = \beta \delta^*(\tilde{q} \mid T)$. Thus,

$$\delta^{\beta Q}(z \mid T) = \sup_{\tilde{q} \in \mathbb{R}^{S^*}} \{\beta(\tilde{q}'z - \delta^*(\tilde{q} \mid T)) - \delta(\tilde{q} \mid Q)\}.$$

Factoring $\beta > 0$ out of the supremum, we get:

$$\delta^{\beta Q}(z \mid T) = \beta \sup_{\tilde{q} \in \mathbb{R}^{S^*}} \{\tilde{q}'z - \delta^*(\tilde{q} \mid T) - \delta(\tilde{q} \mid Q)\} = \beta \delta^Q(z \mid T).$$

2.5 Normalization

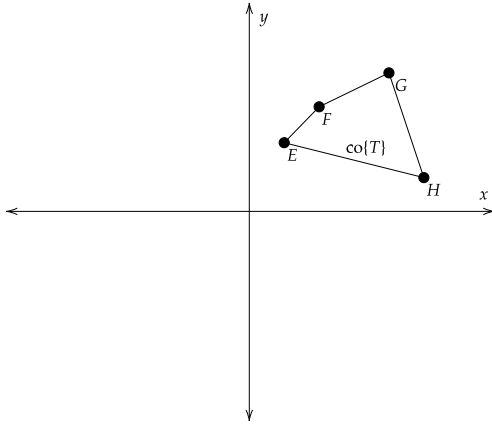
We choose the normalization set Q inside the polar cone C of the recession cone defined in Section 2.3. Let $C = (T_\infty)^\circ$, and take $Q \subset C$ so that $\delta^*(q \mid T)$ is finite. Intuitively, normalization selects Q and thereby fixes the direction (numeraire) in which the inefficiency of z is measured relative to T .

- (a) Input normalization, $Q^x = \{q \in C : q_x \cdot x = -1\}$
- (b) Production normalization, $Q^y = \{q \in C : q_y \cdot y = 1\}$
- (c) Sequestration normalization, $Q^s = \{q \in C : q_s \cdot s = 1\}$
- (d) Abatement normalization, $Q^b = \{q \in C : q_b \cdot b = -1\}$
- (e) Input–Production normalization, $Q^{xy} = \{q \in C : q_x \cdot x = -1, q_y \cdot y = 1\} = Q^x \cap Q^y$
- (f) Input–Production–Sequestration normalization, $Q^{xys} = \{q \in C : q_x \cdot x = -1, q_y \cdot y = 1, q_s \cdot s = 1\} = Q^x \cap Q^y \cap Q^s$
- (g) Input–Production–Abatement normalization, $Q^{xyb} = \{q \in C : q_x \cdot x = -1, q_y \cdot y = 1, q_b \cdot b = -1\} = Q^x \cap Q^y \cap Q^b$
- (h) Integrated normalization, $Q^r = \{q \in C : q_x \cdot x = -1, q_y \cdot y = 1, q_s \cdot s = 1, q_b \cdot b = -1\} = Q^x \cap Q^y \cap Q^s \cap Q^b$

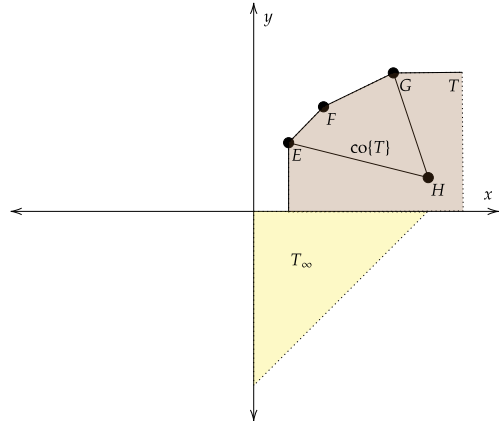
For brevity, we denote the set of all normalizations by

$$\mathcal{Q} = \{Q^x, Q^y, Q^s, Q^b, Q^{xy}, Q^{xys}, Q^{xyb}, Q^r\}.$$

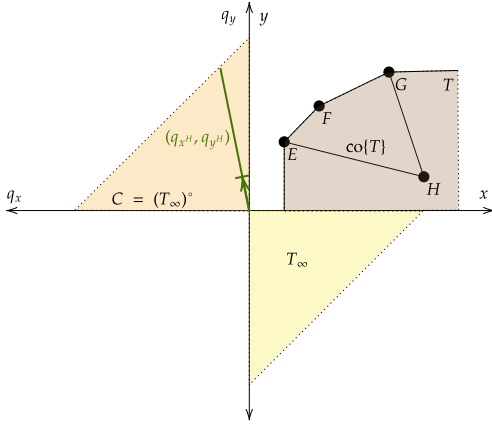
The first four normalizations are pure orientations that focus on single netput types. Q^x measures inefficiency in terms of input savings: how much input cost could be reduced while maintaining



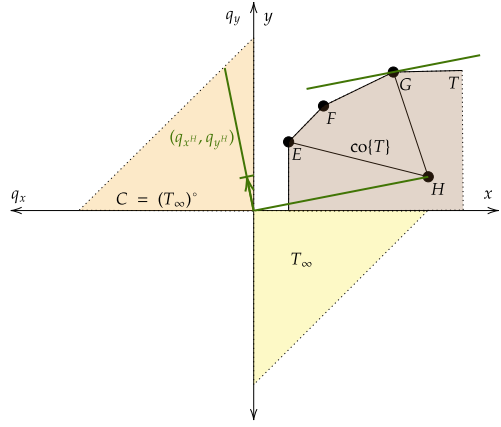
(a) Convex hull $EFGH$



(b) Netput disposability and the recession cone



(c) Polar cone and normalization



(d) The Pareto efficiency measure

Figure 1: Visual illustration of the key geometric concepts leading to the Pareto efficiency measure.

feasibility. Thus, the numeraire is the input bundle normalized as $q_x \cdot x = -1$. Q^y evaluates inefficiency in terms of output gains: how much additional production is feasible given current inputs. The numeraire is the primary output priced so that $q_y \cdot y = 1$. For environmental outcomes, Q^s measures inefficiency as a foregone beneficial environmental output. In contrast, Q^b measures inefficiency by the extent to which adverse environmental outputs could be reduced while remaining within the technology. The numeraire for Q^s is the beneficial environmental output priced as $q_s \cdot s = 1$, and for Q^b , it is the adverse environmental output bundle priced as $q_b \cdot b = -1$.

The last four are mixed normalizations that fix directions for multiple netput types simultaneously and evaluate joint performance across netput dimensions. Q^{xy} measures inefficiency when inputs contract and primary output expands jointly. The numeraire is the pair (x, y) , priced so

that $q_x \cdot x = -1$ and $q_y \cdot y = 1$. Q^{xys} adds beneficial environmental output and interprets the inefficiency as the shortfall in contracting inputs, expanding primary output, and increasing beneficial environmental output. Its numeraire is the triplet (x, y, s) , priced so that $q_x \cdot x = -1$, $q_y \cdot y = 1$, and $q_s \cdot s = 1$. Q^{xyb} pairs input contraction and output expansion with reductions in adverse environmental outputs. The numeraire is (x, y, b) , priced so that $q_x \cdot x = -1$, $q_y \cdot y = 1$, and $q_b \cdot b = -1$. Finally, Q^r sets all four netputs as the unit of account and provides a single integrated index across production and environmental dimensions. Its numeraire is the quadruple (x, y, s, b) , priced so that $q_x \cdot x = -1$, $q_y \cdot y = 1$, $q_s \cdot s = 1$, and $q_b \cdot b = -1$. These normalizations differ only by positive scaling of q and thus preserve the sign of all shadow prices, meaning that the economic interpretation of a positive or negative price is invariant across $Q \in \mathcal{Q}$. We relate the set of normalizations $\mathcal{Q}$ with different policy schemes later in Section 4.

In Figure 1, we visually illustrate the concepts of the convex hull, netput disposability, the recession cone, normalization in the polar cone of the recession, and the efficiency measure in $\mathbb{R}^2$. We consider a simple technology T with a single input x and a single output y , both of which are freely disposable. Let E , F , G , and H denote four observed input–output bundles of T . Figure 1a shows the convex hull $EFGH$ formed by these observed (x, y) points, representing all feasible convex combinations of the data.

Because both x and y are freely disposable, it is always possible to increase x for non-decreasing y , and decrease y for constant x . This property expands the feasible set to include all points weakly dominated by those on the convex hull. Figure 1b illustrates this by adding the recession cone of the technology, T_∞ , shown in yellow, which spans the second quadrant $\{\mathbb{R}_+, \mathbb{R}_-\}$.⁹ The recession cone shows the directions in which the technology is unbounded.

Figure 1c shows the polar cone $C = (T_\infty)^\circ$, which consists of all vectors q that form nonpositive inner products with every direction in the recession cone T_∞ . The polar cone is the space in which we impose our normalization choices, and each $q \in C$ defines a supporting hyperplane to the technology set T . Because T_∞ spans $\{\mathbb{R}_+, \mathbb{R}_-\}$, the polar cone lies in $\{\mathbb{R}_-, \mathbb{R}_+\}$, shown in orange. From the menu of normalizations $\mathcal{Q}$, we consider Q^x , Q^y , and Q^{xy} , each corresponding to a direction in the polar cone along which q is normalized. In this example, we select Q^{xy} to evaluate the observation H : the green line represents this normalization, and the arrow from the origin with a cap along the green line gives the actual dual vector for H , with the cap representing the dual point for H derived as $(q_{x^H}, q_{y^H}) = \left(-\frac{1}{x}, \frac{1}{y}\right)$.

Figure 1d illustrates the efficiency measure. The green lines represent supporting hyperplanes to the technology set in the primal space (the first quadrant) given (q_{x^H}, q_{y^H}) . The maximal supporting hyperplane consistent with feasibility passes through G , a point on the frontier. The

⁹The recession cone $T_\infty \subset \{\mathbb{R}_+, \mathbb{R}_-\}$ allows unbounded movement in the negative y -direction, which is standard in netput representations where outputs may be used as inputs. In our single-output setting, y remains positive, so this is a slight misuse of notation that we adopt for consistency with a canonical technology.

efficiency measure is the algebraic gap between the two hyperplanes, given by $q^\top H - q^\top G$.

2.6 Polyhedral Approximation and Normalizations

Each normalization defines a primal program and a corresponding dual. Table 2 lists the primal and dual formulations for all benchmark normalizations. In each primal, the efficiency measure is expressed as the minimal adjustment needed to move an observed field to the boundary of the frontier. The scaling variables $(\alpha_x, \alpha_y, \alpha_s, \alpha_b)$ represent contractions or expansions of their respective netputs, and the weights γ_i select convex combinations of observed fields that construct the piecewise-linear frontier. The dual gives the same measure in terms of shadow values derived using marginal values q . Each set of dual variables (q_x, q_y, q_s, q_b) assigns implicit values to inputs, grain production, and environmental outcomes consistent with a supporting hyperplane of $T^\mathcal{D}$. Thus, the primal–dual pairs show two complementary views of efficiency: geometric adjustment factors in the primal and the corresponding system of valuations in the dual.

2.7 Environmental Shadow Prices

We follow Podinovski (2019) to compute shadow prices for environmental outputs. Shadow prices are identified by selecting the supporting hyperplane to the production frontier that gives the marginal rate of transformation between an environmental output and grain production.¹⁰ For sequestration s , this corresponds to minimizing $(-q_s)$, and for each adverse environmental output b_m , it corresponds to maximizing q_{b_m} , with $q_y = 1$ so that all shadow prices are expressed in bushels of grain per unit of environmental output.

For $\mathcal{Q} = \{Q^x, Q^y, Q^s, Q^b, Q^{xy}, Q^{xys}, Q^{xyb}, Q^r\}$, the shadow prices for s and b_m are:

$$\partial_s \delta^\mathcal{Q}(z|T^\mathcal{D}) = \min_{q, \phi} \left\{ \begin{array}{l} -q_s : \\ q_y = 1, \\ obj(\delta^\mathcal{Q}) = \delta^{\mathcal{Q}*}, \\ \text{All other dual feasibility constraints of } \delta^\mathcal{Q} \end{array} \right\} \quad (9)$$

$$\partial_{b_m} \delta^\mathcal{Q}(z|T^\mathcal{D}) = \max_{q, \phi} \left\{ \begin{array}{l} q_{b_m} : \\ q_y = 1, \\ obj(\delta^\mathcal{Q}) = \delta^{\mathcal{Q}*}, \\ \text{All other dual feasibility constraints of } \delta^\mathcal{Q} \end{array} \right\}. \quad (10)$$

¹⁰Our identification of shadow prices is structural. They are recovered from the convex geometry of observed netput bundles under assumptions of convexity and disposability, with dual normalizations fixing the valuation.

Table 2: Primal and dual formulations of the efficiency measure δ^Q under different normalizations.

Primal	Dual
$\delta^{Q^x} \min_{\alpha_x, \gamma} \left\{ \begin{array}{l} \alpha_x : \alpha_x x_k^1 \geq \sum_i \gamma_i x_{ik}^1, \alpha_x x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ y \leq \sum_i \gamma_i y_i, s = \sum_i \gamma_i s_i, b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \& \alpha_x \in [0, 1] \end{array} \right\} - 1$	$\max_{q, \phi_x} \left\{ \begin{array}{l} \sum_k q_{xk} x_k - \phi_x : \sum_{k=1}^7 q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m \\ \leq \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_m q_{bm} b_{im}, \forall i = 1, \dots, I, \\ \sum_k q_{xk} x_k - \phi_x = 1, q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_x, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R} \end{array} \right\} - 1$
$\delta^{Q^y} \max_{\alpha_y, \gamma} 1 - \left\{ \begin{array}{l} \alpha_y : x_k^1 \geq \sum_i \gamma_i x_{ik}^1, x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ \alpha_y y \leq \sum_i \gamma_i y_i, s = \sum_i \gamma_i s_i, b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \& \alpha_y \geq 1 \end{array} \right\}$	$\min_{q, \phi_y} 1 - \left\{ \begin{array}{l} q_y y - \phi_y : \sum_{k=1}^7 q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_{m=1}^3 q_{bm} b_{im} \\ \leq \sum_{k=1}^7 q_{xk} x_k - q_y y - q_s s + \sum_{m=1}^3 q_{bm} b_m, \forall i = 1, \dots, I, \\ q_y y - \phi_y = 1, q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_y, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R} \end{array} \right\}$
$\delta^{Q^s} \max_{\alpha_s, \gamma} 1 - \left\{ \begin{array}{l} \alpha_s : x_k^1 \geq \sum_i \gamma_i x_{ik}^1, x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ y \leq \sum_i \gamma_i y_i, \alpha_s s = \sum_i \gamma_i s_i, b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \& \alpha_s \geq 1 \end{array} \right\}$	$\min_{q, \phi_s} 1 - \left\{ \begin{array}{l} q_s s - \phi_s : \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_m q_{bm} b_{im} \\ \leq \sum_k q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m, \forall i = 1, \dots, I, \\ \sum_{l=1}^2 q_{sl} s_l - \phi_s = 1, q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_s, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R} \end{array} \right\}$
$\delta^{Q^s} \min_{\alpha_b, \gamma} \left\{ \begin{array}{l} \alpha_b : \alpha_x x_k^1 \geq \sum_i \gamma_i x_{ik}^1, \alpha_x x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ y \leq \sum_i \gamma_i y_i, s = \sum_i \gamma_i s_i, \alpha_b b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \& \alpha_b \in [0, 1] \end{array} \right\} - 1$	$\max_{q, \phi_b} \left\{ \begin{array}{l} \sum_m q_{bm} b_m - \phi_b : \sum_k q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m \\ \leq \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_m q_{bm} b_{im}, \forall i = 1, \dots, I, \\ \sum_m q_{bm} b_m - \phi_b = 1, q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_b, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R} \end{array} \right\} - 1$
$\delta^{Q^{xy}} \min_{\alpha, \gamma} \left\{ \begin{array}{l} -\alpha_y + \alpha_x : \alpha_x x_k^1 \geq \sum_i \gamma_i x_{ik}^1, \alpha_x x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ \alpha_y y \leq \sum_i \gamma_i y_i, s = \sum_i \gamma_i s_i, b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \alpha_y \geq 1 \& \alpha_x \in [0, 1] \end{array} \right\}$	$\max_{q, \phi} \left\{ \begin{array}{l} q_x x - q_y y - \phi_x + \phi_y : \sum_k q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m \\ \leq \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_{m=1}^3 q_{bm} b_{im}, \forall i = 1, \dots, I, \\ \sum_k q_{xk} x_k - \phi_x = 1, q_y y - \phi_y = 1, \\ q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_x, \phi_y, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R}. \end{array} \right\}$
$\delta^{Q^{xys}} \min_{\alpha, \gamma} 1 + \left\{ \begin{array}{l} -\alpha_y - \alpha_s + \alpha_x : \\ \alpha_x x_k^1 \geq \sum_i \gamma_i x_{ik}^1, \alpha_x x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ \alpha_y y \leq \sum_i \gamma_i y_i, \alpha_s s = \sum_i \gamma_i s_i, \\ b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \alpha_y, \alpha_s \geq 1 \& \alpha_x \in [0, 1] \end{array} \right\}$	$\max_{q, \phi} 1 + \left\{ \begin{array}{l} \sum_k q_{xk} x_k - q_y y - q_s s - \phi_x + \phi_y + \phi_s : \\ \sum_k q_{xk} x_k - q_y y - q_s s + \sum_{m=1}^3 q_{bm} b_m \\ \leq \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_m q_{bm} b_{im}, \forall i = 1, \dots, I, \\ \sum_k q_{xk} x_k - \phi_x = 1, q_y y - \phi_y = 1, \sum_l q_{sl} s_l - \phi_s = 1, \\ q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_x, \phi_y, \phi_s, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R}. \end{array} \right\}$
$\delta^{Q^{xyb}} \min_{\alpha, \gamma} \left\{ \begin{array}{l} -\alpha_y + \alpha_x + \alpha_b : \\ \alpha_x x_k^1 \geq \sum_i \gamma_i x_{ik}^1, \alpha_x x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ \alpha_y y \leq \sum_i \gamma_i y_i, s = \sum_i \gamma_i s_i, \\ \alpha_b b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \alpha_y \geq 1 \& \alpha_x, \alpha_b \in [0, 1] \end{array} \right\} - 1$	$\max_{q, \phi} \left\{ \begin{array}{l} \sum_k q_{xk} x_k - q_y y + \sum_m q_{bm} b_m - \phi_x + \phi_y - \phi_b : \\ \sum_k q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m \\ \leq \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_m q_{bm} b_{im}, \forall i = 1, \dots, I, \\ \sum_k q_{xk} x_k - \phi_x = 1, q_y y - \phi_y = 1, \sum_m q_{bm} b_m - \phi_b = 1, \\ q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_x, \phi_y, \phi_b, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R}. \end{array} \right\} - 1$
$\delta^{Q^r} \min_{\alpha, \gamma} \left\{ \begin{array}{l} -\alpha_y - \alpha_s + \alpha_x + \alpha_b : \\ \alpha_x x_k^1 \geq \sum_i \gamma_i x_{ik}^1, \alpha_x x_k^2 = \sum_i \gamma_i x_{ik}^2, \\ \alpha_y y \leq \sum_i \gamma_i y_i, \alpha_s s = \sum_i \gamma_i s_i, \alpha_b b_m = \sum_i \gamma_i b_{im}, \\ \gamma_i \geq 0, \sum_i \gamma_i = 1, \alpha_y, \alpha_s \geq 1 \& \alpha_x, \alpha_b \in [0, 1] \end{array} \right\}$	$\max_{q, \phi} \left\{ \begin{array}{l} \sum_k q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m - \phi_x + \phi_y + \phi_s - \phi_b : \\ \sum_k q_{xk} x_k - q_y y - q_s s + \sum_m q_{bm} b_m \\ \leq \sum_k q_{xk} x_{ik} - q_y y_i - q_s s_i + \sum_m q_{bm} b_{im}, \forall i = 1, \dots, I, \\ \sum_k q_{xk} x_k - \phi_x = 1, q_y y - \phi_y = 1, q_s s - \phi_s = 1, \\ \sum_m q_{bm} b_m - \phi_b = 1, \\ q_{x_k^1} \geq 0, q_{x_k^2} \in \mathbb{R}, \phi_x, \phi_y, \phi_s, \phi_b, q_y \geq 0, q_s \& q_{bm} \in \mathbb{R}. \end{array} \right\}$

Here $obj(\delta^Q)$ denotes the objective function of program δ^Q , and δ^{Q*} is its optimal value. The equality constraint $obj(\delta^Q) = \delta^{Q*}$ ensures the shadow prices are computed at a supporting hyperplane that defines the frontier.

These formulations are consistent with interpreting shadow prices as one-sided directional derivatives of the conjugate function, as described in (Podinovski, 2019), and have a straightforward economic interpretation. Linear program (9) gives the opportunity cost in bushels of grain for a one-unit increase in sequestration s , and (10) gives the opportunity cost in bushels for a one-unit reduction in an adverse environmental output b_m .

We convert the shadow price from bushels per unit of environmental output to USD per acre by multiplying it by the grain price and scaling by the ratio of the environmental variable to land area. Let p_y denote the grain price (USD per bushel), then the shadow price for s is

$$-q_s^{\text{USD/acre}} = -q_s^{\text{bushels/unit of } s} \times p_y \times \frac{s}{x_1}$$

and for b_m is

$$-q_{b_m}^{\text{USD/acre}} = -q_{b_m}^{\text{bushels/unit of } b_m} \times p_y \times \frac{b_m}{x_1}.$$

Linear programs (9) and (10) can be unbounded or infeasible. Unboundedness arises when no finite supporting hyperplane exists under the chosen normalization. In economic terms, the local trade-off is infinite at the evaluation point. Infeasibility arises when the required marginal change (more s or less b) lies outside the convex hull implied by weak disposability of most netputs, or when sign or normalization restrictions are incompatible with any convex combination of the data. In both cases, shadow prices are undefined (Podinovski & Førsund, 2010; Podinovski, 2019) and, as a consequence, the number of fields for which we obtain shadow prices is smaller than the total sample. To stabilize point estimates, we resample our fields 200 times with replacement and rerun (9) and (10) to retrieve shadow prices for s and b . Pooling these replicates yields 200 times the number of fields in each tillage type, minus those lost to unbounded or infeasible solutions. We trim extremes using the 3.5 *Median Absolute Deviation* rule.

3 Data

Our sample comprises 1,125 corn fields with 85,099 acres of farmland in 53 Illinois counties. We obtain our dataset from Precision Conservation Management (PCM) and the National Oceanic and Atmospheric Administration (NOAA). PCM is a farmer-led initiative helping farmers develop financially viable conservation strategies to address environmental issues (PCM, 2024). PCM collects field-level data on input and output quantities, costs, and returns from participating growers in

Illinois. PCM collaborates with Cool Farm Tool (CFT) and Field to Market (FTM) to provide estimates of sustainability metrics and facilitate farmers’ participation in ecosystem services markets. We use soil conservation metrics (water and wind erosion) from FTM and carbon sequestration and GHG emissions measures from CFT.

CFT provides estimates of carbon sequestration and GHG emissions at the field level consistent with methodologies recommended by the Intergovernmental Panel on Climate Change (IPCC).¹¹ It distinguishes between direct emissions, such as those arising from soil processes and on-farm activities, and indirect upstream emissions embodied in the manufacture and transport of inputs. We include only direct emissions, such as N₂O emissions from fertilizer applications and residue decomposition, CO₂ from fuel use in machinery, and from pesticide applications. CFT estimates carbon sequestration by modeling changes in soil organic carbon stocks, using field-specific characteristics such as soil type, residue retention, and crop yields to quantify annual removals of atmospheric CO₂ into soils and biomass.

FTM provides estimates of soil conservation outcomes based on established erosion models developed by the U.S. Department of Agriculture (USDA) and the Natural Resources Conservation Service (NRCS).¹² Water erosion is calculated using the Revised Universal Soil Loss Equation (RUSLE2) and the Water Erosion Prediction Project (WEPP), while wind erosion is estimated with the Wind Erosion Equation (WEQ) and the Wind Erosion Prediction System (WEPS). These process-based models combine information on soil erodibility, slope, rainfall, crop cover, and management practices to generate annual soil loss per acre.

Daily precipitation data are obtained at the county level from the NOAA for 2022 and 2023 (NOAA, 2024). We calculate the total precipitation for the crop year 2023 by summing daily precipitation amounts spanning October 2022 to September 2023. While the corn production period primarily occurs from April to September, including the full year’s precipitation data is critical for assessing environmental variables. As discussed, SDII is a significant factor for environmental outcomes, which we calculate as

$$\text{SDII} = \frac{\text{sum of daily precipitation in wet days}}{\text{number of wet days}} \quad (11)$$

We present netput summary statistics in Table 3 on a per-acre basis.¹³ The mean corn production in our sample is 225 bushels per acre (SD = 25.8), while the median is 230 bushels per acre. Our sample mean is higher than the Illinois state average yield of 206 bushels per acre in 2023 (USDA NASS, 2024). Carbon sequestration has a mean of 0.13 tCO₂ eq./acre (SD = 0.10) and a

¹¹For more details on carbon metrics, see <https://coolfarm.org/the-tool/>.

¹²For more details on Field to Market metrics, see <https://fieldtomarket.org/national-indicators-report/soil-conservation/>.

¹³Values are reported on a per-acre basis for comparability across fields, although the analysis is conducted using field-level data. Table A1 in Appendix A provides the corresponding summary statistics for the field-level values.

Table 3: Summary Statistics of Netputs Used in the Crop Production Technology on a Per-acre Basis.

Netputs	Mean	SD	Median	p25	p75
Production (lbs./acre)	225.48	25.81	230.00	212.00	244.00
Sequestration (tCO ₂ eq./acre)	0.13	0.10	0.08	0.07	0.16
GHG emissions (tCO ₂ eq./acre)	0.28	0.03	0.27	0.25	0.30
Water erosion (tons of soil/acre)	0.03	0.09	0.01	0.01	0.03
Wind erosion (tons of soil/acre)	0.01	0.04	0.00	0.00	0.00
Land (Acre)	75.64	50.41	68.00	39.21	94.84
Soil Productivity Index (SPI)	131.84	11.65	136.12	125.47	141.23
Nonland costs (\$/acre)	705.70	80.05	702.73	655.42	767.87
Soil erodibility factor	0.31	0.07	0.32	0.24	0.37
Slope (%)	1.75	2.31	0.90	0.50	2.50
Precipitation (mm)	855.20	100.72	822.13	814.87	936.78
Simple Daily Intensity Index (SDII)	5.56	1.64	5.57	3.95	6.85

Note: Parentheses after the netputs provide the units for the statistics. SD denotes standard deviation. p25 and p75 denote the first and third quartiles. SPI and SDII denote the Soil Productivity Index and the Simple Daily Intensity Index, respectively. All summary statistics for wind erosion are zero due to the use of two decimal places in the formatting. The full values are: p25 = 0.00004, median = 0.00016, and p75 = 0.00290.

median of 0.08 tCO₂ eq./acre. Greenhouse gas (GHG) emissions average 0.28 tCO₂ eq./acre (SD = 0.03), with a median of 0.27 tCO₂ eq./acre. Water erosion averages 0.03 tons of soil per acre (SD = 0.09), with a median of 0.01 tons. Wind erosion has a mean of 0.01 tons per acre (SD = 0.04). Although the table displays zeros due to rounding, the underlying values are very small: 0.00004 at the 25th percentile, 0.00016 at the median, and 0.00290 at the 75th percentile. [USDA \(2020\)](#) report that sheet and rill erosion on cropland in Illinois was 4.31 tons per acre in 2017, while wind erosion averaged less than 0.01 tons per acre. The lower water erosion in our sample reflects the flatter topography of our sample fields compared to the statewide average.

The mean field size in our sample is 75.6 acres (SD = 50.4), and the median is 68 acres with an interquartile range of 55.63 acres. The mean soil productivity index (SPI) in our sample is 131.8 (SD = 11.7), with a median of 136.1 (IQR = 15.7). This places most of our fields within the *Class B* to *Class A* range under Illinois soil classification, which includes SPI values between 117 and 147 for prime agricultural soils ([Olson, 2000](#)). The mean nonland cost in our sample is \$705.7 per acre with a standard deviation of \$80.1 per acre. The mean soil erodibility factor in our sample is 0.31 (SD = 0.07), which lies near the middle of the statewide range of 0.02 to 0.69 reported for Illinois soils ([Goldman et al., 1986](#)). The sample slope averages 1.75 percent (SD = 2.31). The distribution is skewed toward flatter fields, with a median of 0.90 percent, though the upper quartile reaches 2.5 percent. Under NRCS slope classes, most fields qualify as nearly level (0–2 percent), with a portion extending into the gently sloping range (2–5 percent) ([Schoeneberger et al., 2017](#)).

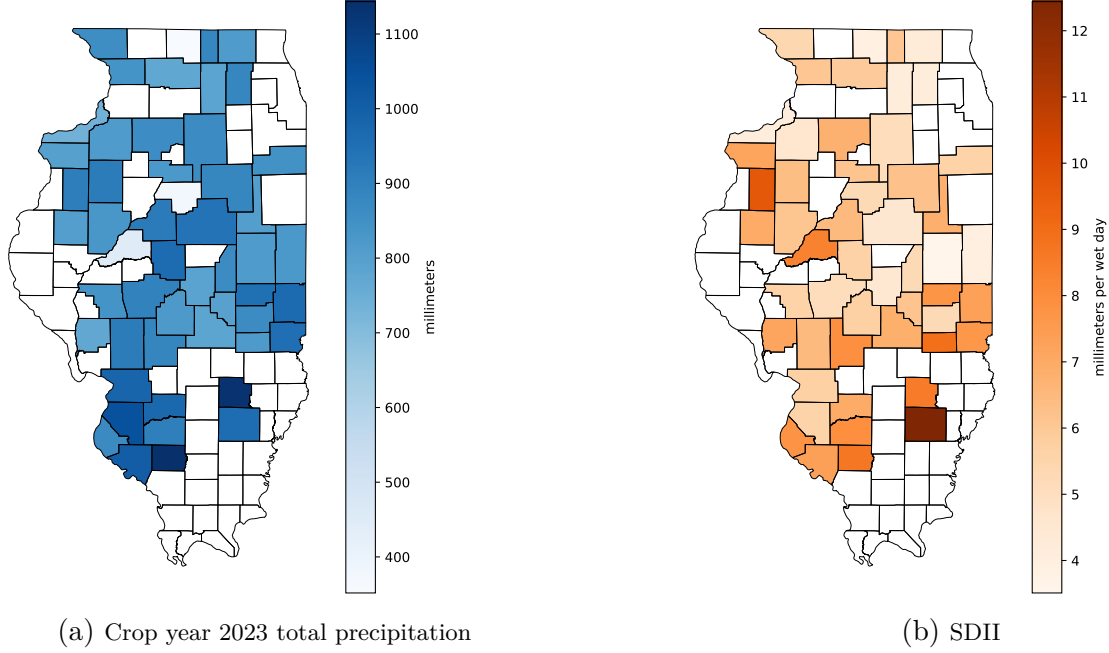


Figure 2: Crop year 2023 total precipitation and simple daily intensity index (SDII) by Illinois county

The total precipitation in crop year 2023 averages 855 mm ($SD = 101$), with a median of 822 mm. The SDII has a mean of 5.56 mm/day ($SD = 1.64$) and a median of 5.57 mm/day. Figure 2 maps total precipitation for crop year 2023 and the SDII across Illinois counties in which our sample fields are located. Precipitation is higher in central and southern Illinois and lower in the northern part of the state. SDII exhibits less spatial contrast, although higher values are present in parts of southern Illinois, and lower values are observed in northern counties. This pattern suggests that while total water availability varies substantially across the state, rainfall intensity is more uniformly distributed, with most counties experiencing similar event intensities.

To approximate our polyhedral technology, we median-normalize netputs to reduce dispersion in magnitudes and improve computational efficiency (Sarkis, 2007). We median-normalize each netput using equation (12).

$$\text{netput}^{\text{norm}} = \frac{\text{netput}}{\text{median}\{\text{netput}\}} \quad (12)$$

4 Policy schemes and the normalization

Shadow prices derived under alternative normalizations Q in the set $\mathcal{Q}$ reveal how the marginal costs or benefits of environmental services vary with farmers' optimization objectives—whether focusing on cost minimization, revenue maximization, environmental outcomes, or combinations thereof. By

mirroring actual policy structures, the framework enables consistent comparison of how alternative programs affect private costs and environmental outcomes.

Table 4 establishes the parallel between different payment policy schemes and our normalizations with examples of existing programs. Conservation programs fall into two main types: practice-based and outcome-based.¹⁴ Practice-based programs reward farmers for implementing specific conservation practices, such as cover cropping or conservation tillage. The Environmental Quality Incentives Program (EQIP) and the Conservation Stewardship Program (CSP) exemplify this approach, providing multi-year payments to encourage sustained implementation of conservation practices (Cooper & Signorello, 2008; Park et al., 2023). Because payments are tied to practices rather than measured environmental outcomes, these programs are simpler to administer and do not require environmental monitoring. Since they do not alter market prices for outputs or inputs, farmers continue to optimize production and cost decisions based on market incentives—increasing grain production, reducing costly inputs, or both. This behavior corresponds to normalizations Q^x , Q^y , or Q^{xy} .

Outcome-based programs reward farmers based on the quantity of environmental benefits achieved, similar to the Payments for Ecosystem Services, which target carbon storage, GHG reduction, and soil loss prevention (Liu & Ruebeck, 2020; Pinto-Correia et al., 2022). These schemes effectively introduce prices for environmental netputs, aligning with normalizations Q^s , Q^b , Q^{xys} , Q^{xyb} , and Q^r in our framework. Normalizations Q^s and Q^b focus on environmental outputs while ignoring grain production, making them unrealistic for commercial grain producers (we include them for analytical completeness). More realistic outcome-based objectives combine environmental goals with economic considerations through joint normalizations: Q^{xys} , Q^{xyb} , and Q^r . The Q^{xys} normalization aligns with the pay-for-carbon programs like the Carbon by Indigo program (Indigo Ag, 2025), while Q^{xyb} corresponds to pay-for-abatement programs that compensate farmers for reducing adverse environmental outputs. Our comprehensive normalization Q^r reflects the multi-objective programs that reward farmers for overall improvements in ecosystem services.

Studies have found that outcome-based payments can deliver more environmental improvement per dollar because they tie compensation to actual results rather than prescribed practices (Bartkowski et al., 2019; OECD, 2022). However, agricultural pollution is a non-point source; its outcomes are therefore diffused and difficult to measure. Monitoring, validating, and crediting outcomes add costs and create payment uncertainty, which can reduce farmer participation. Given these measurement and administrative challenges, practice-based programs remain popular despite lower environmental returns per dollar because they are easier to implement.

Our connection between normalizations and policy instruments is a stylized analogy, not a structural equivalence. Normalizations are grounded in production theory and duality, while policy

¹⁴In the literature and in practice, practice-based programs are also referred to as input-, action-, or measure-based, while outcome-based programs are called results-, output-, success-, or objective-based payments.

Table 4: Mapping of conservation payment policy schemes to corresponding normalizations.

Policy scheme	Normalizations $\mathcal{Q}$	Numeraire	Examples
Practice-based payment	Q^x Q^y Q^{xy}	x y (x, y)	Environmental Quality Incentives Program (EQIP, 2025), Conservation Stewardship Program (CSP, 2025), Bayer Carbon (Bayer, 2025) and Agoro Carbon Programs (Agoro, 2025)
Outcome-based payment			
<i>Pay-for-carbon</i>	Q^s Q^{xys}	s (x, y, s)	Hypothetical; not realistic Carbon by Indigo (Indigo Ag, 2025)
<i>Pay-for-abatement</i>	Q^b Q^{xyb}	b (x, y, b)	Hypothetical; not realistic Regrow programs for carbon reduction (Regrow Ag, 2025) and Ohio River Basin Water Quality Trading Project (EPRI, 2025)
<i>Pay-for-carbon and abatement</i>	Q^r	(x, y, s, b)	Ecosystem Services Market Consortium (ESMC, 2025) and Soil and Water Outcomes Fund (2025)

rules emerge from institutional, political, and contractual considerations. Regulators do not choose normalizations in the mathematical sense; rather, they design incentive structures that these normalizations help us understand. The mapping reveals how different normalizations parallel the incentive structures that policy schemes generate, demonstrating how alternative normalizations can illuminate the types of trade-offs that policymakers seek to structure and evaluate.

5 Environmental Services Shadow Prices

We present two complementary analyses of environmental shadow prices. First, we examine how shadow prices vary across normalizations using the pooled sample of all fields. This reveals how alternative policy frameworks, practice-based versus outcome-based, affect the valuation of environmental services when applied to the same production technology. Second, we disaggregate the sample by tillage type and compare shadow prices between conservation and conventional tillage systems under each normalization. This comparison identifies the marginal incentive payments required to encourage the adoption of conservation tillage and demonstrates how these costs vary systematically across different policy designs. Together, these analyses show that both the policy framework (normalization) and management practice (tillage type) jointly determine the private costs of environmental compliance.

Table 5 presents summary statistics of shadow prices for environmental outputs (in USD per acre) for the pooled sample of all fields, obtained from linear programs (9) and (10). Shadow price signs indicate the direction of adjustment required in the numeraire. For carbon sequestration s ,

negative values indicate costs—storing an additional ton of CO₂ requires either increasing valued inputs or foregoing crop revenue. Positive values indicate synergies where sequestration can be increased while reducing input costs or expanding output. For adverse environmental outputs b , the interpretation reverses: positive values indicate abatement costs (requiring more inputs or producing less output). In contrast, negative values signal opportunities to reduce environmental damage while saving on inputs or increasing output. These sign conventions remain consistent across all normalizations $\mathcal{Q}$, and changing the normalization rescales the magnitudes but does not alter the interpretation of signs.

Median carbon sequestration shadow prices are negative across all normalizations, representing the marginal costs of increasing carbon sequestration. The median costs vary significantly by optimization approach and follow a clear pattern: normalizations aligned with practice-based policies (Q^x , Q^y , and Q^{xy}) yield the lowest costs. In contrast, those aligned with outcome-based policies (Q^s , Q^{xys} , and Q^r) yield the highest costs. Normalization Q^{xy} produces the minimum shadow price of USD -20.33 per acre, indicating that farmers optimizing input use and corn production face relatively low costs for increasing carbon sequestration. The maximum median cost is under Q^{xys} at USD -372.74 per acre, where farmers maximize sequestration and crop production, while minimizing production inputs. This high cost reflects the difficulty of further increasing sequestration when it is already prioritized alongside competing objectives. Q^{xyb} and Q^b are notable exceptions—despite being aligned with outcome-based policies, they produce costs similar to practice-based approaches because sequestration is not in their numeraire; therefore, the trade-off is measured relative to improving input use and corn production while reducing adverse environmental outputs.

For GHG emissions, median shadow prices are positive under every normalization, representing marginal abatement costs. Similar to sequestration, median costs are low for normalizations aligned with practice-based policies (Q^x , Q^y , and Q^{xy}), ranging from USD 411.26 to 977.50 per acre, and high for those aligned with outcome-based policies (Q^{xyb} , Q^{xys} , Q^r , Q^s , and Q^b), ranging from USD 1057.92 to 1703.87. The highest median cost is under Q^b at USD 1703.87 per acre, consistent with increasing marginal costs of abatement¹⁵, followed by Q^s , Q^r , Q^{xys} , and Q^{xyb} .

Median shadow prices for water erosion are positive across all normalizations, indicating the costs of erosion abatement. The highest median cost is under Q^b at USD 159.26 per acre, while the lowest is under Q^{xy} at USD 44.90 per acre. For wind erosion, median shadow prices are also positive across all normalizations, ranging from USD 39.06 under Q^{xy} to USD 91.71 under Q^{xys} . In contrast to sequestration and GHG emissions, erosion control exhibits similar costs across normalizations, with variation substantially smaller than for carbon or emissions outcomes. This pattern suggests that trade-offs between corn production and erosion are less sensitive to the choice of numeraire.

Figure 3 visually summarizes the median shadow prices from Table 5. Shadow prices are color-

¹⁵ Q^b is unrealistic as it assumes fields target only emission abatement.

Table 5: Summary statistics of shadow prices for environmental outputs in USD per acre under normalizations $\mathcal{Q}$

	p25	p50	p75
Q^x			
s	-72.59	-27.04	12.62
b_1	-107.39	411.26	959.75
b_2	-24.73	72.04	227.35
b_3	15.40	50.32	116.12
Q^y			
s	-61.75	-21.37	8.53
b_1	516.83	977.50	1469.55
b_2	-32.02	57.42	169.32
b_3	13.87	40.38	85.96
Q^{xy}			
s	-56.77	-20.33	8.08
b_1	369.63	682.89	1049.71
b_2	-26.29	44.90	150.87
b_3	12.42	39.06	88.10
Q^s			
s	-306.85	-141.18	-64.73
b_1	588.86	1295.30	2193.85
b_2	-81.46	62.65	230.10
b_3	11.71	45.86	111.92
Q^b			
s	-73.20	-30.23	5.91
b_1	1159.75	1703.87	2434.49
b_2	-22.73	159.26	399.19
b_3	19.33	53.29	115.99
Q^{xys}			
s	-1111.91	-372.74	-98.02
b_1	509.69	1141.63	1905.77
b_2	-131.72	67.04	341.05
b_3	21.59	91.71	296.55
Q^{xyb}			
s	-61.03	-24.76	5.42
b_1	839.17	1057.92	1322.88
b_2	-13.19	87.32	213.46
b_3	16.33	46.18	91.28
Q^r			
s	-960.44	-216.84	-78.75
b_1	713.98	1270.67	2069.20
b_2	-156.47	55.89	289.87
b_3	16.54	71.79	214.48

Note: s , b_1 , b_2 , and b_3 denote carbon sequestration, GHG emissions, water erosion, and wind erosion. p25, p50, and p75 denote first, second (median), and third quartiles.

coded with blues representing negative values and oranges representing positive values. For both colors, darker shades represent more extreme values and lighter shades represent values closer to zero. The figure is divided into left and right quadrants aligned with practice-based and outcome-based policies. Carbon sequestration and GHG emissions show differences between policy approaches, with darker shades concentrated on the right quadrant. The figure reveals that GHG emissions

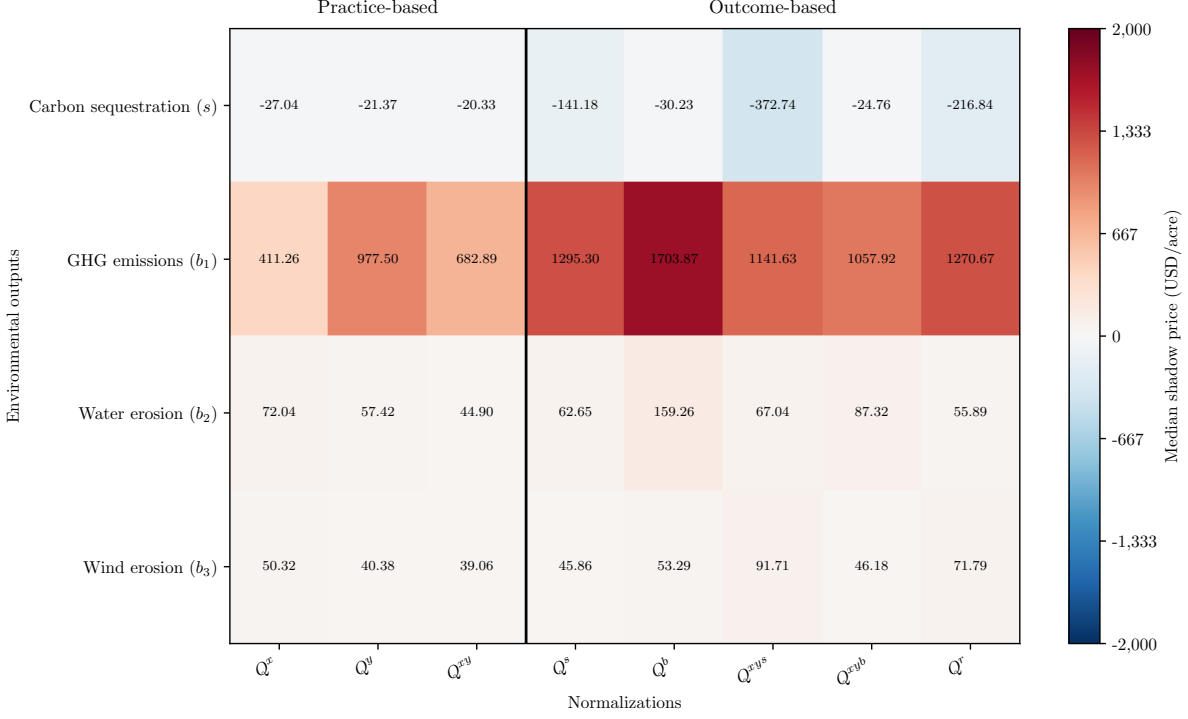


Figure 3: Distribution of median shadow prices (USD per acre) for environmental outputs under normalizations Q

abatement is by far the most expensive environmental service for our sample fields, especially under outcome-based interventions. At the opposite extreme, carbon sequestration under practice-based normalizations shows the lowest costs. However, carbon sequestration presents substantially larger costs under outcome-based interventions, highlighting the relevance of carefully planning policy objectives. As discussed, the relatively light and homogeneous palette for erosion control suggests low costs across normalizations, demonstrating that these outcomes are less sensitive to the choice of normalization.

The analysis thus far has shown that normalizations aligned with practice-based and outcome-based policies produce systematically different shadow price estimates. This raises a natural question: when designing incentive programs to encourage conservation practices, how do these policy-dependent valuations translate into differential costs across management systems?

To answer this, we compare shadow prices between conservation and conventional tillage fields under each normalization. These differences quantify the marginal incentive payments required to encourage the adoption of conservation tillage and reveal how costs vary depending on whether the policy emphasizes input efficiency, output production, or environmental outcomes. All results are expressed on a per-acre basis to facilitate comparison across incentive types.

We focus on conservation versus conventional tillage practices.¹⁶ Shadow prices are calculated for both management systems under all normalizations, with their difference representing the marginal incentive required to shift from conventional to conservation tillage. We first summarize the differences between the two tillage systems, then present the estimated shadow prices and derived incentive payments. Table A2 in Appendix A reports summary statistics of netputs by tillage type.

Conservation tillage involves minimal soil disturbance, while conventional tillage involves extensive disturbance through plowing and inversion (Baker et al., 2007). Conservation tillage includes no-till, strip-till, and mulch-till systems that maintain at least 30% crop residue after planting, whereas conventional tillage involves full-width tillage that leaves less than 15% residue (CTIC, 2002; Claassen et al., 2018). These practices affect soil structure, moisture retention, and nutrient cycling, which influence production netputs.

By selecting suitable tillage methods, growers can enhance conditions for higher yields while minimizing adverse environmental impacts. Studies on conservation tillage report enhanced soil fertility and improved crop yields. Retained crop residues act as mulch that conserves moisture and returns nutrients to the soil, supporting crop growth (Derpsch et al., 2010; Kassam et al., 2015; Holland, 2004). However, a global meta-analysis by Pittelkow et al. (2015) finds that, under certain conditions, conservation tillage can result in lower yields than conventional tillage, suggesting that the optimal strategy varies by soil type, climate, and crop. Conventional tillage may initially improve aeration and weed control, but it tends to degrade soil structure and reduce water retention over time, ultimately diminishing long-term productivity (Baker et al., 2007).

Conservation tillage enhances carbon sequestration and reduces GHG emissions. By minimizing disturbance and maintaining residue cover, it promotes the accumulation of soil organic carbon, which is important for capturing atmospheric CO₂ and reducing GHG emissions (Ogle et al., 2005; Paustian et al., 2016). Conversely, conventional tillage increases GHG emissions by disturbing the soil, disrupting the soil matrix, accelerating organic-matter decomposition, and releasing GHGs to the atmosphere (Reicosky, 1997). This reduces the soil’s capacity to act as a carbon sink, amplifying the climate impact of tillage (Paustian et al., 2019).

Conservation tillage mitigates water and wind erosion by maintaining soil structure and residue cover. This protects soil particles from detachment and transport by water and wind. Residue cover reduces raindrop splash, slows runoff, increases infiltration, and strengthens soil structure, maintaining physical integrity against erosive forces (Montgomery, 2007). In contrast, conventional tillage exacerbates erosion by leaving soil bare and susceptible to water and wind. Intensive tillage disrupts soil aggregates, reduces organic matter, and lowers resistance to erosion, increasing soil degradation and loss (Pimentel et al., 1995).

¹⁶While cover cropping is another relevant conservation practice in the US, insufficient observations in our sample (less than 20% of fields) preclude its analysis.

Table 6: Summary statistics of environmental output shadow prices in USD per acre by tillage type under normalizations $\mathcal{Q}$.

	Conservation tillage fields			Conventional tillage fields		
	p25	p50	p75	p25	p50	p75
Q^x						
s_1	-94.69	-40.47	-4.65	-120.23	-62.36	0.24
b_1	-211.62	450.92	980.89	188.13	617.62	1151.94
b_2	-66.49	29.62	154.42	-14.24	75.91	345.93
b_3	26.50	57.54	126.43	5.78	45.88	112.75
Q^y						
s_1	-89.17	-50.52	-26.61	-81.77	-41.32	3.37
b_1	580.83	1188.34	1670.88	665.30	1169.72	1648.02
b_2	-20.31	66.38	209.62	-47.91	48.92	193.04
b_3	28.56	55.80	99.76	-0.78	25.10	65.78
Q^{xy}						
s_1	-76.83	-43.29	-15.75	-79.67	-42.68	3.50
b_1	444.46	799.96	1228.50	506.54	788.76	1195.35
b_2	-28.79	50.97	173.71	-29.68	40.08	161.36
b_3	23.54	52.42	96.06	0.69	28.23	71.12
Q^{xys}						
s_1	-1151.50	-710.33	-178.87	-1136.80	-399.94	-124.32
b_1	648.36	1313.55	2034.63	735.17	1209.91	1933.75
b_2	-34.32	110.27	433.88	-50.26	174.47	506.17
b_3	39.90	102.83	244.62	6.10	68.44	239.57
Q^{xyb}						
s_1	-84.83	-50.05	-22.64	-87.21	-48.52	-3.73
b_1	908.93	1140.17	1463.62	906.43	1160.31	1410.67
b_2	-31.44	65.51	228.53	-30.18	77.18	266.46
b_3	31.73	60.24	103.19	3.54	31.21	72.79
Q^r						
s_1	-1116.58	-491.10	-161.22	-1053.50	-264.78	-106.75
b_1	859.37	1379.25	2090.30	806.04	1278.97	2071.31
b_2	-48.04	116.90	416.86	-85.45	150.68	504.83
b_3	36.44	83.32	192.47	2.65	59.75	196.08

Note: s , b_1 , b_2 , and b_3 denote carbon sequestration, GHG emissions, water erosion, and wind erosion. p25, p50, and p75 denote first, second (median), and third quartiles.

We derive environmental shadow prices using (9) and (10), separately by tillage type, and present results in Table 6.¹⁷ The tillage-specific shadow prices follow the same qualitative patterns as the pooled sample. Carbon sequestration median values are negative across all normalizations, while the medians for GHG emissions and erosion (both water and wind) remain positive under both tillage systems, confirming that the basic trade-offs are consistent across management approaches. For carbon sequestration and GHG emissions, the relative ranking across normalizations mirrors the pooled results: costs are generally lower under normalizations aligned with practice-based policies and higher under normalizations aligned with outcome-based policies.

In contrast to the pooled sample, the costs of abating water erosion show a clearer separation

¹⁷We do not report Q^s and Q^b because, as discussed, these normalization extremes are unrealistic for policy design.

between practice-based and outcome-based policy-aligned normalizations. Median costs are lower under Q^x , Q^y , and Q^{xy} and higher under Q^{xys} and Q^r . This clearer separation reflects heterogeneity in erosion abatement costs across tillage systems that is masked in the pooled analysis.

For wind erosion, the tillage-specific results do not show the clearer separation seen for water erosion. This indicates that trade-offs between production and wind erosion are less sensitive to the choice of normalization, even when fields are separated by tillage type.

We compare the marginal costs of providing environmental services between conservation and conventional tillage using the difference in their median values and present results in Table 7. We use 10,000 bootstrap resamples of shadow prices for each tillage type to quantify sampling uncertainty around these differences. For each resample, we compute group medians and their difference. The 2.5th and 97.5th percentiles of these differences form the 95% confidence intervals, shown in parentheses in Table 7. Confidence intervals that exclude zero indicate statistically significant differences at the 5% level.

For carbon sequestration, a positive value means that conservation tillage delivers the service at a lower marginal cost than conventional tillage. A negative value means that conservation tillage operates at a higher marginal cost, which, under increasing marginal costs, implies a higher level of sequestration service. For adverse environmental outputs, the interpretation reverses.

Table 7: Median Difference in Shadow Prices (USD per acre) with Bootstrapped 95% Confidence Intervals

$\mathcal{Q}$	Δ_s	Δ_{b_1}	Δ_{b_2}	Δ_{b_3}
Q^x	21.89 (19.10, 25.05)	-166.70 (-200.57, -139.30)	-46.29 (-55.48, -38.29)	11.66 (8.67, 15.19)
Q^y	-9.20 (-11.22, -7.12)	18.62 (-6.12, 49.07)	17.46 (10.58, 23.45)	30.70 (28.95, 32.67)
Q^{xy}	-0.61 (-2.15, 1.04)	11.20 (-4.82, 30.73)	10.89 (6.13, 15.96)	24.19 (22.56, 26.09)
Q^{xys}	-310.39 (-352.98, -260.77)	103.64 (71.76, 130.5)	-64.20 (-79.73, -52.26)	34.39 (28.25, 40.02)
Q^{xyb}	-1.54 (-3.11, -0.03)	-20.14 (-32.58, -7.76)	-11.67 (-17.21, -6.27)	29.03 (26.72, 30.97)
Q^r	-226.33 (-256.47, -203.33)	100.28 (71.63, 132.36)	-33.78 (-47.49, -20.96)	23.57 (19.38, 27.9)

Note: $\mathcal{Q}$ denotes normalization. Δ represents the median difference in shadow prices (USD per acre) between conservation and conventional tillage systems. Values in parentheses are the 2.5th and 97.5th percentiles of the bootstrapped distribution of the median difference ($n = 10,000$ iterations). A negative value for carbon sequestration or a positive value for adverse environmental outputs indicates a higher marginal cost for conservation tillage compared to conventional tillage.

Carbon sequestration differences vary in magnitude across normalizations, suggesting that conservation tillage has higher marginal costs under all cases except for Q^x . Under this normalization, conservation tillage fields, which focus on minimizing input use (reducing tillage operations and

thus energy use), provide cheaper sequestration services than conventional tillage fields. Input use minimization may reduce yields in conservation tillage fields, thereby reducing both sequestration potential and its marginal costs. The conservation tillage advantage disappears once output maximization is part of the normalization. Under Q^y and Q^{xy} , conservation tillage no longer offers a cheaper alternative to increase sequestration. Under Q^{xys} and Q^r , the differential costs are larger, indicating that incentives needed to adopt conservation tillage are generally greater under outcome-based schemes. Compensation under Q^{xyb} is more aligned with practice-based cases, as it focuses on mitigating adverse environmental outcomes rather than increasing sequestration.

The difference in marginal costs for GHG emissions between the two tillage systems varies in both magnitude and sign across normalizations in a similar fashion to sequestration. The differences are negative under Q^x , meaning conservation tillage faces lower marginal costs to reduce GHG when fields minimize inputs. The differences become positive once output maximization is incorporated into the normalization (Q^y and Q^{xy}), and they drastically increase under outcome-based schemes (Q^{xys} and Q^r). These normalizations leading to positive differences involve maximizing both output and sequestration, indicating that conservation tillage fields require incentives to reduce GHG emissions. The need for incentives disappears when fields are compensated for the quantity of adverse environmental outputs reduced (Q^{xyb}); under this scheme, conservation tillage allows for reducing GHG emissions at lower marginal cost.

The values for water erosion are negative under Q^x , Q^{xys} , Q^{xyb} , and Q^r , indicating lower abatement costs for conservation tillage. In contrast, differences are positive under Q^y and Q^{xy} , meaning that conservation tillage fields that focus on maximizing agricultural production while ignoring environmental outcomes, have higher marginal costs and would require compensation to expand water erosion abatement.

The values for wind erosion are positive under all normalizations, ranging from USD 11.66 to 34.39 per acre. This suggests that conventional tillage fields operate at higher baseline levels of wind erosion, making additional abatement relatively inexpensive, while conservation tillage fields have already achieved better erosion outcomes but face higher marginal costs for further improvements.

Across all normalizations, the values in Table 7 are statistically significant, as the bootstrapped 95% confidence intervals exclude zero, with the exceptions of Δs under Q^{xy} and Δb_1 under Q^y and Q^{xy} . To summarize, our results suggest that conservation tillage entails higher marginal costs for carbon sequestration and wind erosion, but its relative costs for GHG abatement and water erosion depend on the normalization used. The fact that Q^{xyb} does not usually behave like outcome-based schemes and is associated with lower relative costs highlights the importance of not generalizing the results across different types of compensation schemes.

6 Conclusions

We develop a duality-based framework for valuing agricultural environmental services under policy-consistent normalizations. A normalization is a formal choice of numeraire that embeds policy objectives into the efficiency and valuation process. We consider practice-based programs that reward farmers for implementing specific conservation practices, such as cover cropping or conservation tillage, and outcome-based programs that reward farmers based on the environmental benefits achieved. By linking normalizations to the design of conservation payment programs, the framework connects production theory with policy evaluation, showing how shadow prices depend on the perspective from which they are derived.

Empirical results from Illinois corn production demonstrate that shadow prices for environmental services vary with normalization. Practice-based normalizations yield lower shadow prices for carbon sequestration and greenhouse gas abatement than outcome-based normalizations. Erosion control costs remain stable across normalizations, suggesting that trade-offs between crop production and erosion are less affected by policy design. When fields are grouped by tillage type, conservation tillage shows higher marginal costs for sequestration and wind erosion but lower or comparable costs for greenhouse gas and water erosion abatement, depending on the normalization applied.

These findings suggest that shadow prices are not fixed technological parameters, but rather policy-dependent measures. The duality structure clarifies this relationship: each normalization alters the direction in which the technology is evaluated, changing how trade-offs among production, sequestration, emissions, and erosion are measured. The framework offers a consistent basis for comparing conservation strategies and designing incentive programs that align private management with public environmental goals.

References

- Afriat, S. N. (1972). Efficiency estimation of production functions. *International Economic Review*, 568–598.
- Agoro. (2025). *Agoro carbon alliance soil carbon program 2025*. <https://agorocarbonalliance.com/1/soil-carbon-program-2025/>.
- Aigner, D., Lovell, C. K., & Schmidt, P. (1977). Formulation and estimation of stochastic frontier production function models. *Journal of econometrics*, 6(1), 21–37.
- Baker, C. J., Saxton, K. E., Ritchie, W., Chamen, W., Reicosky, D., Ribeiro, M., ... others (2007). *No-tillage seeding in conservation agriculture*. Published jointly by Food and Agriculture Organization of the United Nations
- Banker, R. D., Charnes, A., & Cooper, W. W. (1984). Some models for estimating technical and scale inefficiencies in data envelopment analysis. *Management science*, 30(9), 1078–1092.
- Barrett, C. B. (1996). On price risk and the inverse farm size-productivity relationship. *Journal of Development Economics*, 51(2), 193–215.
- Barrett, C. B., Bellemare, M. F., & Hou, J. Y. (2010). Reconsidering conventional explanations of the inverse productivity–size relationship. *World development*, 38(1), 88–97.
- Bartkowski, B., Droste, N., Ließ, M., Sidemo-Holm, W., Weller, U., & Brady, M. V. (2019). Implementing result-based agri-environmental payments by means of modelling. *arXiv preprint arXiv:1908.08219*.
- Bayer. (2025). *Bayer ForGround carbon initiative*. Web page. Bayer. Retrieved from <https://bayerforground.com/carbon-initiative>
- Burton, R. J., & Schwarz, G. (2013). Result-oriented agri-environmental schemes in europe and their potential for promoting behavioural change. *Land use policy*, 30(1), 628–641.
- Chambers, R. G. (2025). *Conjugate paretian inefficiency measures*. (Working Paper)
- Chambers, R. G., Chung, Y., & Färe, R. (1998). Profit, directional distance functions, and nerlovian efficiency. *Journal of optimization theory and applications*, 98, 351–364.
- Chambers, R. G., & Färe, R. (2020). Distance functions in production economics. In S. C. Ray, R. G. Chambers, & S. C. Kumbhakar (Eds.), *Handbook of production economics* (p. 1-35). Springer Singapore.
- Charnes, A., Cooper, W. W., Golany, B., Seiford, L., & Stutz, J. (1985). Foundations of data envelopment analysis for pareto-koopmans efficient empirical production functions. *Journal of econometrics*, 30(1-2), 91–107.

- Charnes, A., Cooper, W. W., & Rhodes, E. (1978). Measuring the efficiency of decision making units. *European journal of operational research*, 2(6), 429–444.
- Choi, Y., Ma, Y., Zhao, Y., & Lee, H. (2023). Inequality in fossil fuel power plants in china: a perspective of efficiency and abatement cost. *Sustainability*, 15(5), 4365.
- Claassen, R., Bowman, M., McFadden, J., Smith, D., & Wallander, S. (2018). *Tillage intensity and conservation cropping in the united states*.
- Claassen, R., & Ribaud, M. (2016). Cost-effective conservation programs for sustaining environmental quality. *Choices*, 31(3), 1–12.
- Cook, W. D., & Seiford, L. M. (2009). Data envelopment analysis (dea)—thirty years on. *European journal of operational research*, 192(1), 1–17.
- Cooper, J., & Signorello, G. (2008). Farmer premiums for the voluntary adoption of conservation plans. *Journal of Environmental Planning and Management*, 51(1), 1–14. doi: 10.1080/09640560701712234
- Corteva. (2023). *Can Corn Get Too Much Nitrogen?* Retrieved 2023-10-06, from <https://www.corteva.us/Resources/crop-protection/corn/too-much-nitrogen-corn.html>
- CSP. (2025). *Conservation Stewardship Program*. <https://www.nrcs.usda.gov/programs-initiatives/csp-conservation-stewardship-program>.
- CTIC. (2002). *Tillage type definitions*. Conservation Technology Information Center. Retrieved 2025-08-12, from https://www.ctic.org/resource_display/?id=322&title=Tillage+Type+Definitions
- Davidson, E. A., & Kanter, D. (2014). Inventories and scenarios of nitrous oxide emissions. *Environmental Research Letters*, 9(10), 105012.
- Derpsch, R., Friedrich, T., Kassam, A., & Li, H. (2010). Current status of adoption of no-till farming in the world and some of its main benefits. *International journal of Agricultural and biological engineering*, 3(1), 1–25.
- Dixit, P. N., Richter, G. M., Coleman, K., & Collins, A. L. (2023). Bioenergy crop production and carbon sequestration potential under changing climate and land use: A case study in the upper river taw catchment in southwest england. *Science of the Total Environment*, 900, 166390.
- Dollan, I. J., Maggioni, V., Johnston, J., Coelho, G. d. A., & Kinter III, J. L. (2022). Seasonal variability of future extreme precipitation and associated trends across the contiguous us. *Frontiers in Climate*, 4, 954892.
- EPRI. (2025). *Ohio river basis water quality trading project*. <https://wqt.epri.com/>.

- EQIP. (2025). *Environmental Quality Incentives Program*. <https://www.nrcs.usda.gov/programs-initiatives/eqip-environmental-quality-incentives>.
- ERS. (2025). *Conservation programs*. U.S. Department of Agriculture, Economic Research Service. Retrieved from <https://www.ers.usda.gov/topics/natural-resources-environment/conservation-programs/>
- ESMC. (2025). *Ecosystem services market consortium*. Retrieved from <https://ecosystems-services-market.org/>
- Färe, R., & Grosskopf, S. (2005). *New directions: efficiency and productivity* (Vol. 3). Springer Science & Business Media.
- Färe, R., Grosskopf, S., & Lovell, C. K. (1985). *The measurement of efficiency of production* (Vol. 6). Springer Science & Business Media.
- Färe, R., Grosskopf, S., & Whittaker, G. (2013). Directional output distance functions: endogenous directions based on exogenous normalization constraints. *Journal of Productivity Analysis*, 40, 267–269.
- Farrell, M. J. (1957). The measurement of productive efficiency. *Journal of the Royal Statistical Society: Series A (General)*, 120(3), 253–281.
- Goldman, S. J., Jackson, K., & Bursztynsky, T. A. (1986). *Erosion and sediment control handbook*. McGraw Hill.
- Greenstone, M., & Gallagher, J. (2008). Does hazardous waste matter? evidence from the housing market and the superfund program. *The Quarterly Journal of Economics*, 123(3), 951–1003.
- Hellerstein, D. M. (2017). The us conservation reserve program: The evolution of an enrollment mechanism. *Land use policy*, 63, 601–610.
- Holland, J. M. (2004). The environmental consequences of adopting conservation tillage in europe: reviewing the evidence. *Agriculture, ecosystems & environment*, 103(1), 1–25.
- Hosseini-Yekani, S.-A., Tomaczewski, S., & Zander, P. (2024). Model-based valuation of ecosystem services using bio-economic farm models: Insights for designing green tax policies and payment for ecosystem services. *Agriculture*, 15(1), 60.
- Indigo Ag. (2025). *Carbon credits: Earn revenue for sustainable farming practices*. <https://www.indigoag.com/carbon-credits>.
- Johnston, R. J., Boyle, K. J., Adamowicz, W., Bennett, J., Brouwer, R., Cameron, T. A., ... others (2017). Contemporary guidance for stated preference studies. *Journal of the Association of Environmental and Resource Economists*, 4(2), 319–405.

- Kassam, A., Friedrich, T., Derpsch, R., & Kienzle, J. (2015). Overview of the worldwide spread of conservation agriculture. *Field Actions Science Reports. The Journal of Field Actions*, 8.
- Keeler, B. L., Polasky, S., Brauman, K. A., Johnson, K. A., Finlay, J. C., O'Neill, A., ... Dalzell, B. (2012). Linking water quality and well-being for improved assessment and valuation of ecosystem services. *Proceedings of the National Academy of Sciences*, 109(45), 18619–18624.
- Keiser, D. A., & Shapiro, J. S. (2019). Consequences of the clean water act and the demand for water quality. *The Quarterly Journal of Economics*, 134(1), 349–396.
- Kumbhakar, S. C., Wang, H.-J., & Horncastle, A. P. (2015). *A practitioner's guide to stochastic frontier analysis using stata*. Cambridge University Press.
- Kunwar, S. R., Chambers, R. G., Gentry, L. F., & Serra, T. (2025). What are the carbon services from cover-crop adoption worth from farmers' perspective? *American Journal of Agricultural Economics*.
- Laborde, D., Mamun, A., Martin, W., Piñeiro, V., & Vos, R. (2021). Agricultural subsidies and global greenhouse gas emissions. *Nature communications*, 12(1), 2601.
- Lapierre, M., Le Velly, G., Bougherara, D., Préget, R., & Sauquet, A. (2023). Designing agri-environmental schemes to cope with uncertainty. *Ecological Economics*, 203, 107610.
- Liu, H., & Ruebeck, C. (2020). Knowledge spillover and positive environmental externality in agricultural decision making under performance-based payment programs. *Agricultural and Resource Economics Review*, 49(2), 270–290. doi: 10.1017/age.2020.18
- Lubowski, R. N., Plantinga, A. J., & Stavins, R. N. (2006). Land-use change and carbon sinks: econometric estimation of the carbon sequestration supply function. *Journal of Environmental Economics and Management*, 51(2), 135–152.
- MacDonald, J. M. (2020). Tracking the consolidation of us agriculture. *Applied Economic Perspectives and Policy*, 42(3), 361–379.
- Menegat, S., Ledo, A., & Tirado, R. (2022). Greenhouse gas emissions from global production and use of nitrogen synthetic fertilisers in agriculture. *Scientific Reports*, 12(1), 1–13.
- Mitchell, J., & Bubenzer, G. (1980). Soil loss estimation. In M. Kikby & R. Morgan (Eds.), *Soil erosion* (p. 17-62). John Wiley and Sons, Ltd., Hoboken.
- Montgomery, D. R. (2007). Soil erosion and agricultural sustainability. *Proceedings of the National Academy of Sciences*, 104(33), 13268–13272.
- NOAA. (2024). *NOAA Climate Data Online*. <https://www.ncei.noaa.gov/cdo-web/>. (Accessed: 2024-11-24)

- Nordhaus, W. D. (2017). Revisiting the social cost of carbon. *Proceedings of the National Academy of Sciences*, 114(7), 1518–1523.
- OECD. (2022). *Making agri-environmental payments more cost effective*. Paris: OECD Publishing. doi: <https://doi.org/10.1787/4cf10d76-en>
- OECD. (2025). *Subsidies and government support*. Retrieved from <https://www.oecd.org/en/topics/subsidies-and-government-support.html>
- Ogle, S. M., Breidt, F. J., & Paustian, K. (2005). Agricultural management impacts on soil organic carbon storage under moist and dry climatic conditions of temperate and tropical regions. *Biogeochemistry*, 72, 87–121.
- Olson, K. R. (2000). *Optimum crop productivity ratings for illinois soil* [Bulletin 811].
- Park, B., Rejesus, R. M., Aglasan, S., Che, Y., Hagen, S. C., & Salas, W. (2023). Payments from agricultural conservation programs and cover crop adoption. *Applied Economic Perspectives and Policy*, 45(2), 984–1007.
- Paustian, K., Larson, E., Kent, J., Marx, E., & Swan, A. (2019). Soil c sequestration as a biological negative emission strategy. *Frontiers in Climate*, 1, 482133.
- Paustian, K., Lehmann, J., Ogle, S., Reay, D., Robertson, G. P., & Smith, P. (2016). Climate-smart soils. *Nature*, 532(7597), 49–57.
- PCM. (2024). *Precision Conservation Management – A Farm Service Program*. Retrieved 2024-03-19, from <https://www.precisionconservation.org/>
- Pimentel, D., Harvey, C., Resosudarmo, P., Sinclair, K., Kurz, D., McNair, M., ... others (1995). Environmental and economic costs of soil erosion and conservation benefits. *Science*, 267(5201), 1117–1123.
- Pindyck, R. S. (2017). The use and misuse of models for climate policy. *Review of Environmental Economics and Policy*.
- Pinto-Correia, T., Ferraz-de Oliveira, M., Guimarães, M., Sales-Baptista, E., Pinto-Cruz, C., Godinho, C., & Santos, R. (2022). Result-based payments as a tool to preserve the high nature value of complex silvo-pastoral systems: progress toward farm-based indicators. *Ecology and Society*, 27(1). doi: 10.5751/es-12973-270139
- Pittelkow, C. M., Liang, X., Linquist, B. A., Van Groenigen, K. J., Lee, J., Lundy, M. E., ... Van Kessel, C. (2015). Productivity limits and potentials of the principles of conservation agriculture. *Nature*, 517(7534), 365–368.

- Podinovski, V. V. (2019). Direct estimation of marginal characteristics of nonparametric production frontiers in the presence of undesirable outputs. *European Journal of Operational Research*, 279(1), 258–276.
- Podinovski, V. V., & Førsund, F. R. (2010). Differential Characteristics of Efficient Frontiers in Data Envelopment Analysis. *Operations Research*, 58(6), 1743–1754.
- Ray, S. C. (2007). Shadow profit maximization and a measure of overall inefficiency. *Journal of Productivity Analysis*, 27, 231–236.
- Regrow Ag. (2025). *Regrow Ag*. <https://www.regrow.ag/>.
- Reicosky, D. (1997). Tillage-induced co₂ emission from soil. *Nutrient cycling in agroecosystems*, 49, 273–285.
- Roberts, M. J., & Bucholtz, S. (2005). Slippage in the conservation reserve program or spurious correlation? a comment. *American Journal of Agricultural Economics*, 87(1), 244–250.
- Sarkis, J. (2007). Preparing your data for DEA. In J. Zhu & W. D. Cook (Eds.), *Modeling Data Irregularities and Structural Complexities in Data Envelopment Analysis* (pp. 305–320). Springer.
- Schoeneberger, P., Wysocki, D., Busskohl, C., & Libohova, Z. (2017). *Soil survey manual*. Washington, DC: United States Department of Agriculture, Natural Resources Conservation Service.
- Sheng, Y., & Chancellor, W. (2019). Exploring the relationship between farm size and productivity: Evidence from the Australian grains industry. *Food Policy*, 84, 196–204.
- Silva, T. P., Bressiani, D., Ebling, É. D., & Reichert, J. M. (2024). Best management practices to reduce soil erosion and change water balance components in watersheds under grain and dairy production. *International soil and water conservation research*, 12(1), 121–136.
- Sobota, D. J., Compton, J. E., McCrackin, M. L., & Singh, S. (2015). Cost of reactive nitrogen release from human activities to the environment in the United States. *Environmental Research Letters*, 10(2), 025006.
- Soil and Water Outcomes Fund. (2025). *Earn more for your acres*. Retrieved from <https://theoutcomesfund.com/>
- Thangarajan, R., Bolan, N. S., Tian, G., Naidu, R., & Kunhikrishnan, A. (2013). Role of organic amendment application on greenhouse gas emission from soil. *Science of the Total Environment*, 465, 72–96.
- Trimble, S. W., & Crosson, P. (2000). US soil erosion rates—myth and reality. *Science*, 289(5477), 248–250.

- USDA. (2020). *Summary report: 2017 national resources inventory* (Summary Report). Washington, DC: Natural Resources Conservation Service, Washington, DC and Center for Survey Statistics and Methodology, Iowa State University, Ames, Iowa.
- USDA. (2022). RUSLE2 Handbook [Computer software manual]. United States Department of Agriculture. (Accessed: 2025-03-21)
- USDA NASS. (2024). *Illinois crop production 2023 summary / corn yield estimates* (Tech. Rep.). USDA NASS, Heartland Regional Field Office. Retrieved from https://www.nass.usda.gov/Statistics_by_State/Illinois/Publications/Current_News_Release/2024/20240112-IL-Annual-Crop-Production.pdf
- Vargas, V. P., Sangoi, L., Ernani, P. R., Picoli Jr., G. J., & Cantarella, H. (2015). Maize leaf phytotoxicity and grain yield are affected by nitrogen source and application method. *Agronomy Journal*, 107(2), 671-679.
- Yoksoulian, L. (2019, April 30). Excessive rainfall as damaging to corn yield as extreme heat, drought. *News Bureau, University of Illinois*. Retrieved from <https://news.illinois.edu/excessive-rainfall-as-damaging-to-corn-yield-as-extreme-heat-drought/> (Accessed: 2025-03-21)
- Yu, Y., Schneider, U., Yang, S., Becker, A., & Ren, Z. (2020). Evaluating the gpcc full data daily analysis version 2018 through etccdi indices and comparison with station observations over mainland of china. *Theoretical and Applied Climatology*, 142, 835–845. doi: 10.1007/s00704-020-03352-8
- Zhou, P., Zhou, X., & Fan, L. (2014). On estimating shadow prices of undesirable outputs with efficiency models: A literature review. *Applied Energy*, 130, 799–806.

Appendix A Summary statistics

Table A1: Summary Statistics of Netputs Used in the Crop Production Technology at the Field Level.

Netputs	Mean	SD	Median	p25	p75
Production (lbs.)	17413.78	12227.00	15205.68	8670.48	22011.00
Sequestration (tCO ₂ eq.)	10.07	11.27	6.55	3.11	12.90
GHG emissions (tCO ₂ eq.)	21.10	14.61	18.34	10.59	26.37
Water erosion (tons of soil)	1.30	1.93	0.86	0.62	1.21
Wind erosion (tons of soil)	0.78	2.40	0.01	0.00	0.19
Land (Acre)	75.64	50.41	68.00	39.21	94.84
SPI	131.84	11.65	136.12	125.47	141.23
Nonland costs (\$)	53596.28	36276.06	47122.74	27636.40	66597.98
Soil erodibility factor	0.31	0.07	0.32	0.24	0.37
Slope (%)	1.75	2.31	0.90	0.50	2.50
Precipitation (mm)	855.20	100.72	822.13	814.87	936.78
SDII	5.56	1.64	5.57	3.95	6.85

Note: Parentheses after the netputs provide the units for the statistics. p25 and p75 denote the first and third quartiles. SPI and SDII denote the Soil Productivity Index and the Simple Daily Intensity Index, respectively.

Table A2: Summary Statistics of Netputs Used in the Crop Production Technology on a Per-acre Basis by Tillage Type.

Netputs	Conservation tillage fields			Conventional tillage fields		
	Median	p25	p75	Median	p25	p75
Production (lbs./acre)	230.00	212.25	245.00	229.00	211.00	243.00
Sequestration (tCO ₂ eq./acre)	0.16	0.15	0.33	0.08	0.06	0.08
GHG emissions (tCO ₂ eq./acre)	0.27	0.26	0.29	0.27	0.25	0.30
Water erosion (tons of soil/acre)	0.01	0.01	0.02	0.02	0.01	0.03
Wind erosion (tons of soil/acre)	0.00	0.00	0.00	0.00	0.00	0.01
Land (Acre)	71.71	39.91	99.44	63.30	39.00	89.28
SPI	134.68	125.13	140.59	137.15	125.77	141.41
Nonland costs (\$/acre)	691.79	651.10	756.12	716.18	656.43	774.93
Soil erodibility factor	0.32	0.24	0.32	0.32	0.24	0.37
Slope (%)	0.90	0.50	3.00	0.90	0.50	1.00
Precipitation (mm)	822.13	813.55	903.45	862.55	814.87	936.78
SDII	5.23	4.14	6.43	5.73	3.95	7.30

Note: Parentheses after the netputs provide the units for the statistics. p25 and p75 denote the first and third quartiles. SPI and SDII denote the Soil Productivity Index and the Simple Daily Intensity Index, respectively. All summary statistics for wind erosion are zero due to the use of two decimal places in the formatting. The full values for conservation tillage are: median = 0.00007, p25 = 0.00002, and p75 = 0.0002 and for conventional tillage are: median = 0.00008, p25 = 0.00066, and p75 = 0.01304.